



Distillation Process Operation

Réf : OTP-DC

N° Série : 221 05 073

OPERATING MANUAL

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INTRODUCTION

This instruction manual is an original document intended for people working on the unit OTP-DC n°22105073.

It is therefore essential that each person using this unit respects all the instructions given in this document.

This manual describes the various features of the unit and the operating procedures to enable a good understanding of different possibilities.

This document contains the following:

- A detailed description of the unit where each part is indicated and its functions described
- Operating risks
- Operating mode
- Maintenance

A technical file includes all the technical documents of the different components so that operators are able to know the maintenance and the operating conditions determined by the different manufacturers.

The equipment is equipped with a nameplate according to the standards respected in its design, conception, and validation. This nameplate contains information relating to the proper operation of the equipment. Operators must respect them carefully.

If the installation and operation of the equipment are not done by PIGNAT personnel, then these operations should only be done by people who have thoroughly read and understood this operating manual.

The operator must ensure the compliance with the installation conditions and carry out the required commissioning declarations to the competent authorities.

Before each intervention, the operators must read and understand the instruction manual which will be completed by a user manual. This last document will be drafted by the responsible of the unit taking into account the procedures and guidelines specific to the operator's company.

PIGNAT SA cannot be held responsible for any damage due to improper use of the equipment, non-compliance with the instruction, negligence, lack of supervision or maintenance or problems caused by persons other than PIGNAT personnel.

TEACHING OBJECTIVES

For the OTP-DC unit, the teaching objectives are to familiarize the student with the use of sensors, actuators and control regulators in industrial technology, as well as the study of the optimization of different control loops.

This unit helps to show:

- The importance of knowing the equipment, its characteristics and its settings
- The importance of knowing the static gain of each of the loops studied
- The importance of knowing the operating conditions
- The effects of the PID parameters
- The reaction of the system to a disruptive factor
- The interest of using a cascade loop
- ...

The equipment is made up of industrial components which, with the help of the information given in the technical documents, allow a technological study.

The following pages describe the unit, how it is set up, its operating mode and some examples of possible experiments. (These pedagogical studies are given as examples and must be completed by the teacher or supervisor according to the level of knowledge of the users.)

DISTILLATION PROCESS OPERATION

OTP-DC/ N° 22105073

OPERATING MANUAL

1. *CERTIFICATION*

1.1. CERTIFICATION

1.1.1. CE certification

Marking of the unit:

 PIGNAT MADE IN FRANCE 6 RUE CALMETTE BP11 69740 GENAS-FRANCE Tél. (33) 478.905.003 Fax. (33) 478.906.388		 Site WEB : www.pignat.com E mail: pignat@pignat.com	
APPARATUS	CONTROLLED DISTILLATION PILOT PLANT		
MODEL - REF	OTP-DC 8 in		
SERIAL NUMBER	22105073	MANUFACTURING YEAR	21
POWER SUPPLY	3X400V+N+G / 12A		50 Hz
COMPRESSED AIR	6	B	1000 l/h
WATER	3	B	5000 l/h
NITROGEN	1	B	3000 l/h
STEAM	8	B	48 kg/h
VACUUM	YES	NET WEIGHT	4250 kg
SIZE	L 555	W 545	H 425 cm

1.1.2. ATEX Certificate

This unit is not designed to be used in an ATEX zone.

The unit must never be put in an ATEX zone.

No product capable of generating an ATEX, other than those specified in this operating manual, must ever be used with this unit.

1.1.3. Pressure Equipment Directive

The Pressure Equipment Directive (DESP) can be applied if the working pressure of the equipment is greater than +0.5 bar (7.25 psi) relative.

« If the working pressure is less than or equal to 0.5 bar (7.25 psi) relative, the equipment is subject to the rules which apply. »

The equipment works at a pressure less than or equal to 0.5 bar (7.25 psi) relative. Therefore, it is not subject to the DESP directive. Its conception and design, manufacture and verification respect the SEP (Sound Engineering Practice) rules.

PIGNAT cannot be held responsible for any use that does not comply with this instruction manual.

DISTILLATION PROCESS OPERATION

OTP-DC/ N° 22105073

OPERATING MANUAL

2. *THE UNIT*

2.1. DESCRIPTION OF THE UNIT

2.1.1. Operating principle

Distillation is a process including heat and material transfers in order to separate one or several products of relative different volatility.

The separation is carried out by a chaining of vaporizations and condensations. The theoretical base is that a vapour produced from a mixture is in equilibrium with the mixture liquid phase and has generally a higher composition in the most volatile compound.

The separation is carried out in a column (packing and bubble cap tray column) designed to get the best contact and material exchanges between the climbing vapour and the falling liquid into the column. This separation can be done in continuous or discontinuous (batch) process.

We ask the user to respect the instructions detailed in this operating book so that the unit can be operated in safety conditions.

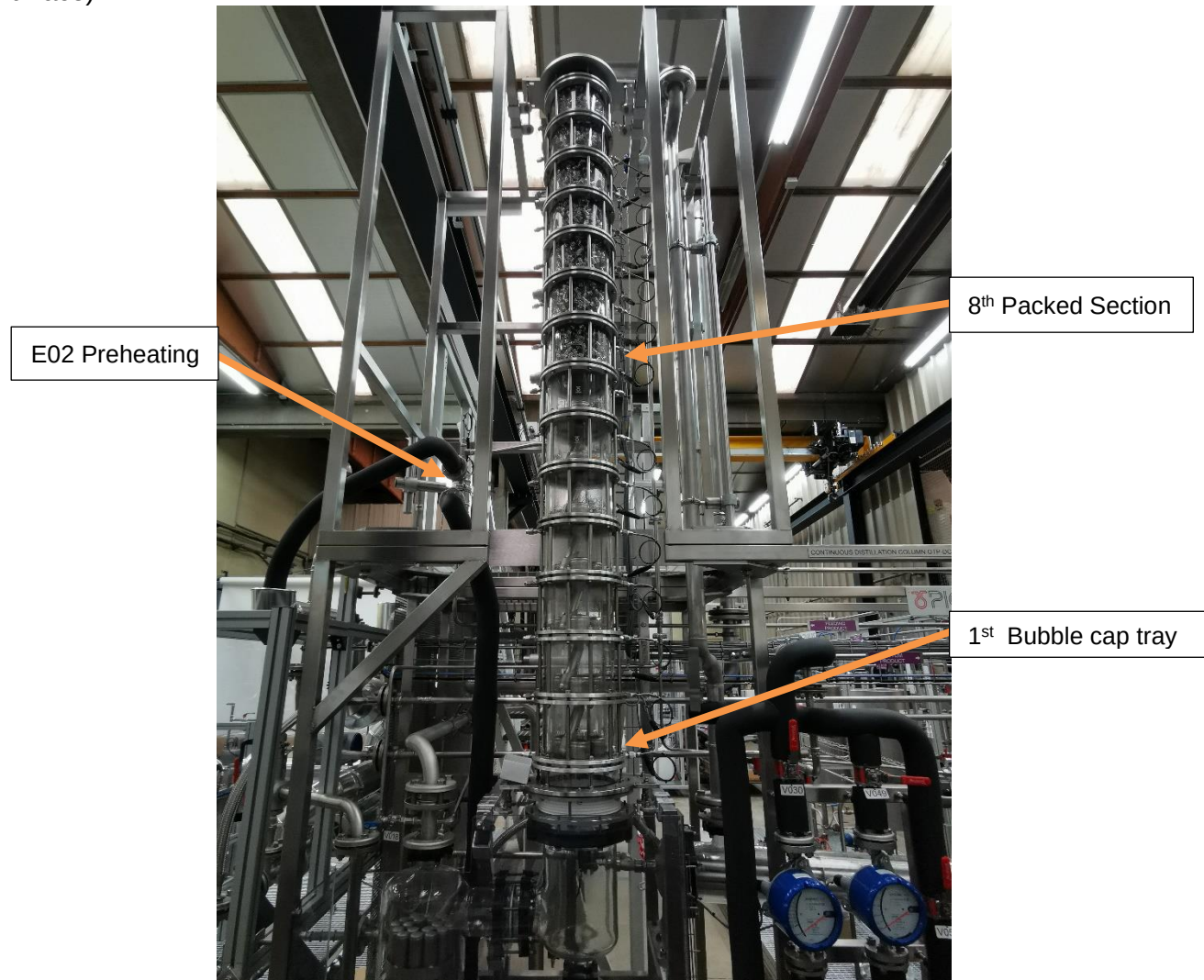
This operating textbook dedicated to the unit should be read carefully by the chemical engineering laboratory manager and by all users of the unit.

You will find in this document:

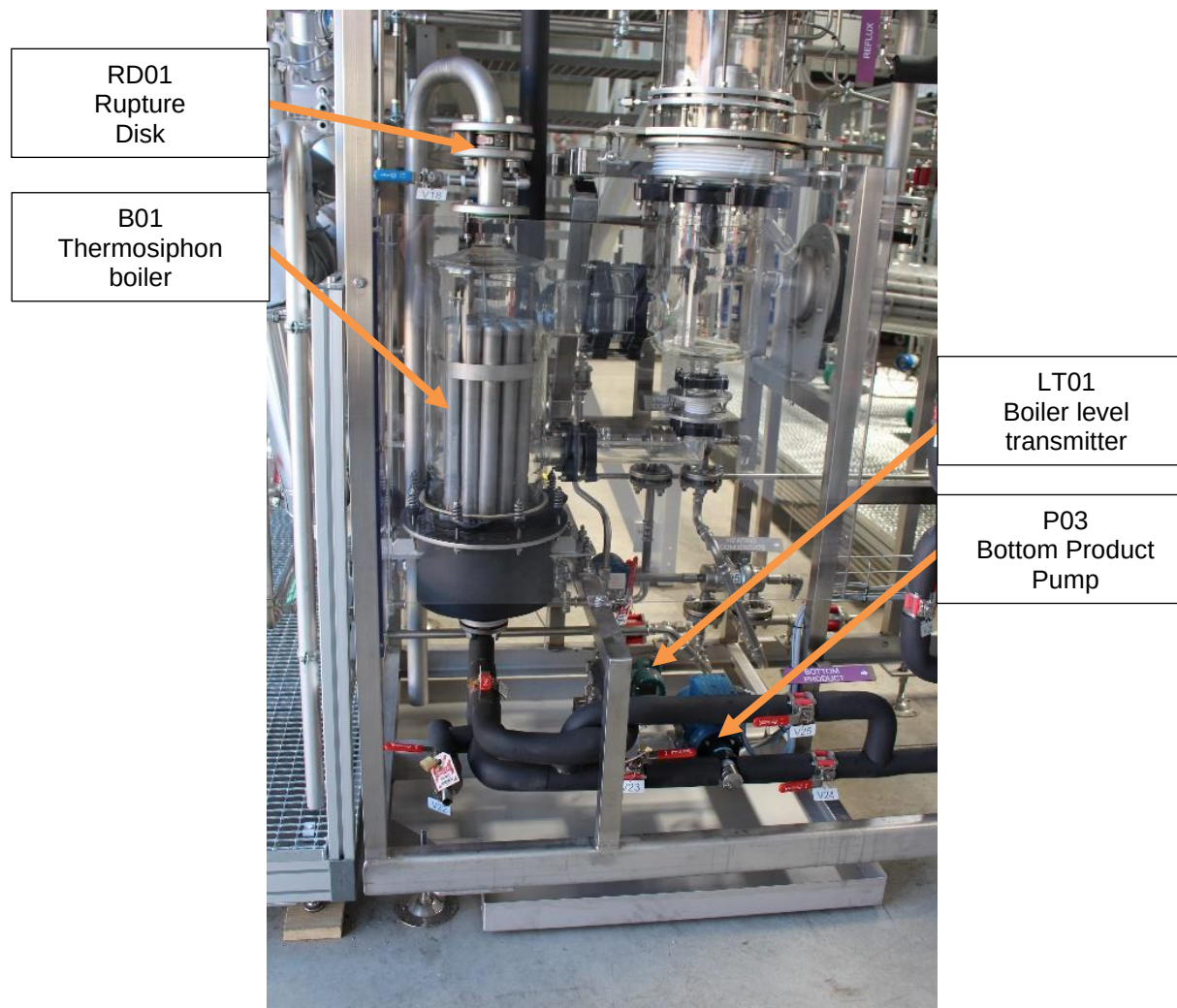
- different functionalities of the control distillation unit
- procedures to operate the unit in safety conditions and in order to make a complete study of the unit and of the distillation process
- safety remarks and information to control all kind of risks while using the distillation unit.

2.1.2. Presentation of the main components

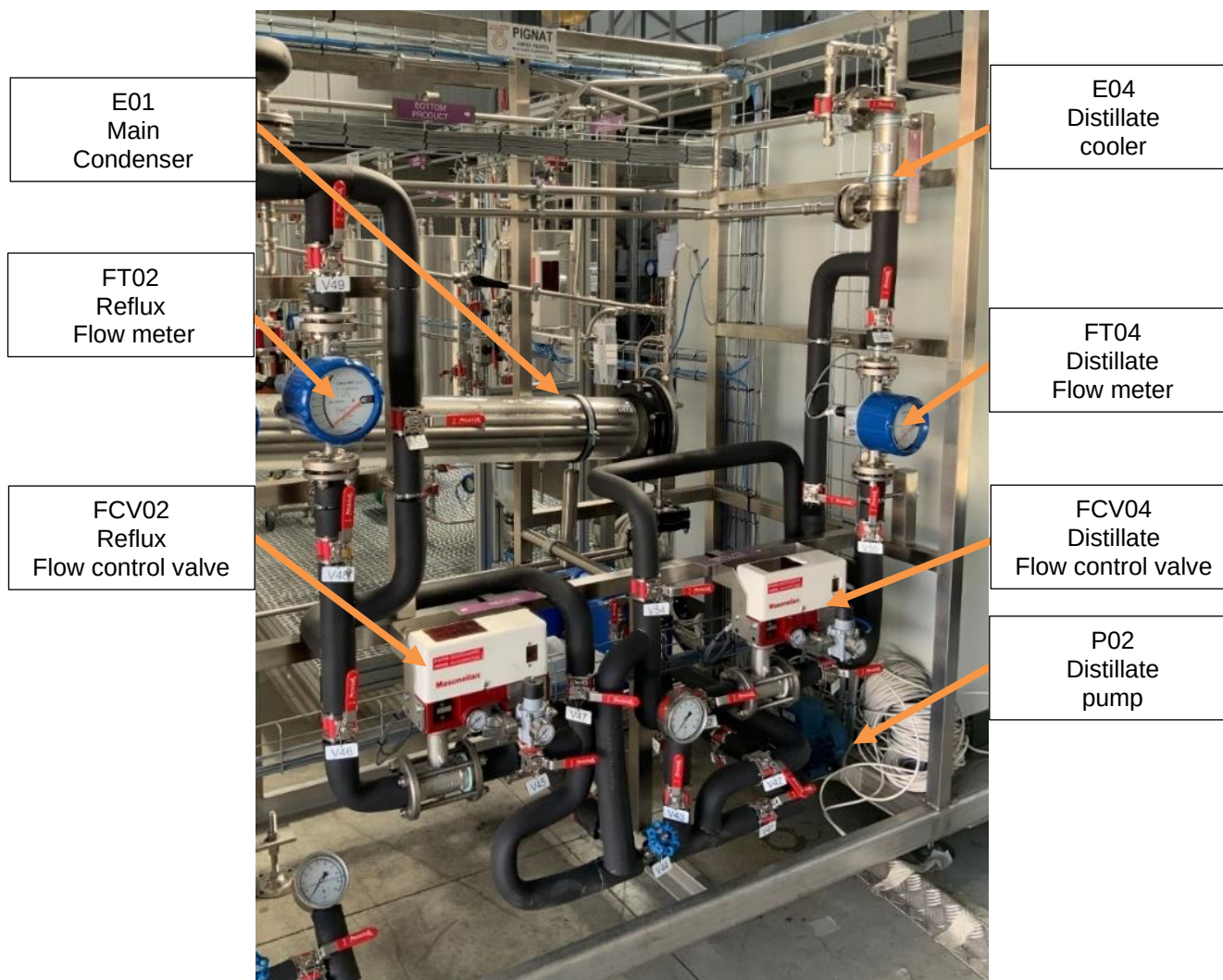
- Column Trays (Front Face):



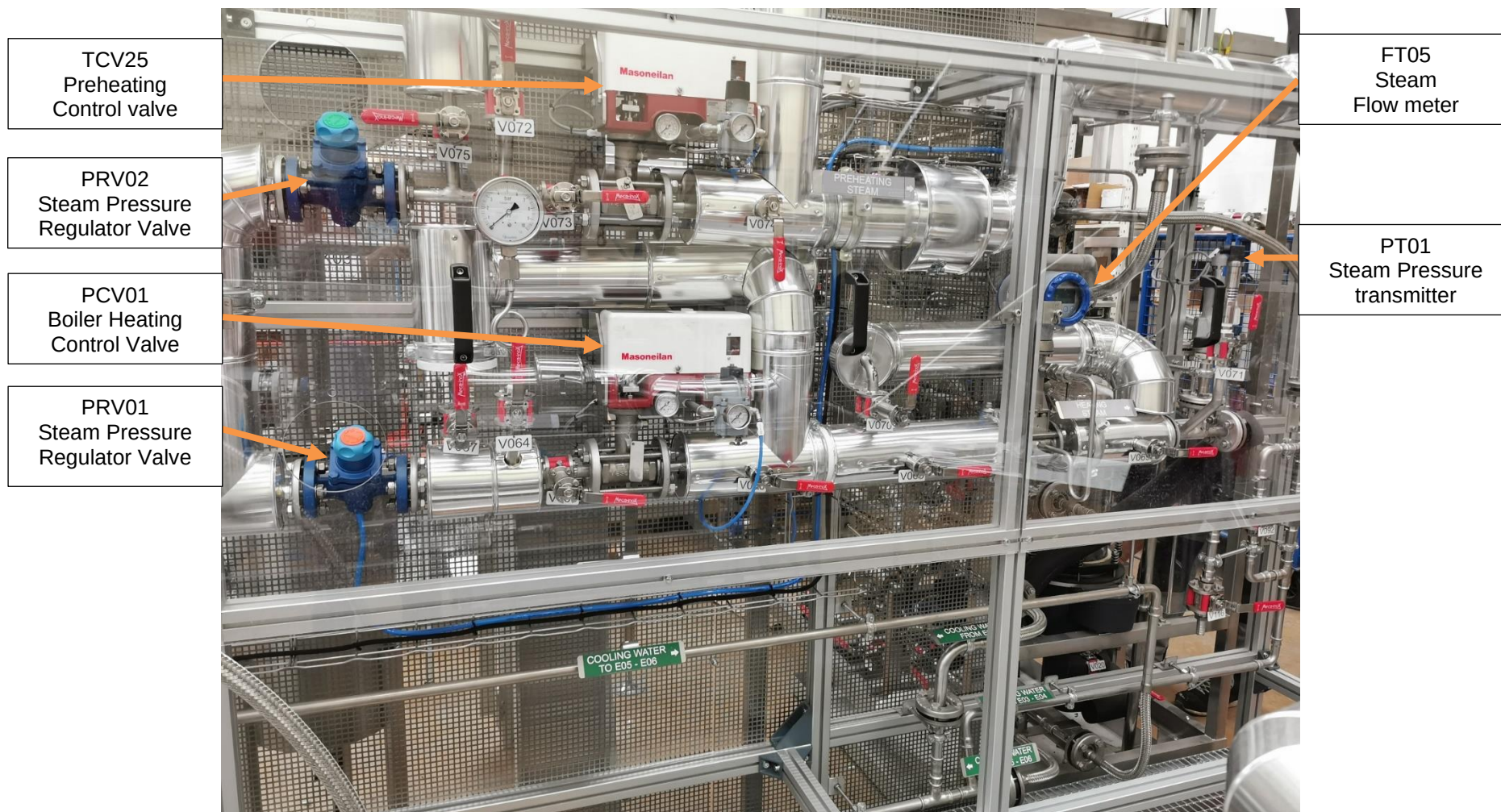
- Boiler (Front Face):



- Distillate/Reflux (Front Face):



- Steam network:



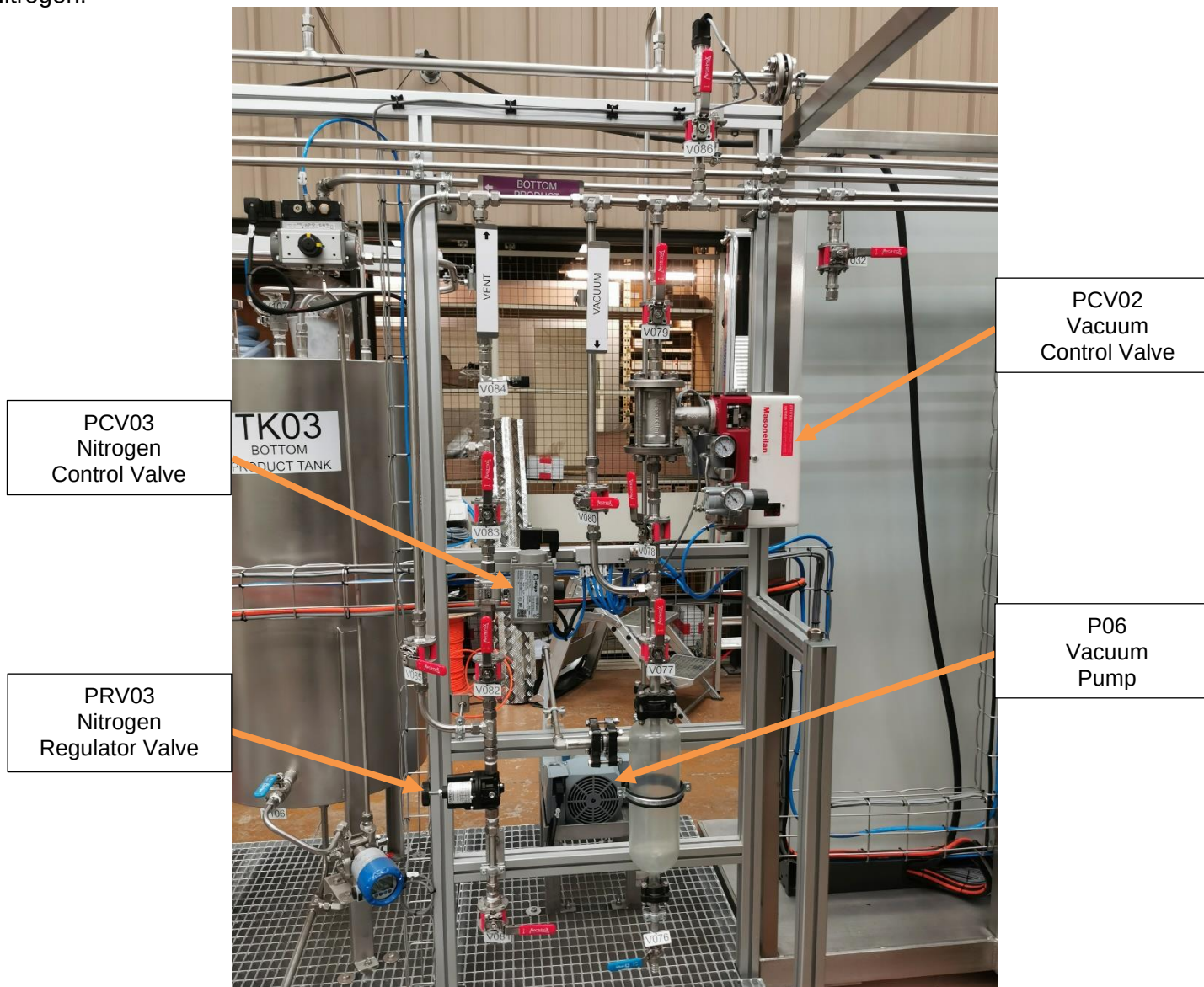
- Cooling water network:

FCV06
Cold Water Flow
Control Valve



FT06
Cold Water
Flow meter

- Vacuum/Nitrogen:



- Crash tank:



From the rupture disk RD01
Extraction to the lab

V120
Draining Valve

- Product Tanks:



2.2. OPERATING RANGE

2.2.1. Operating conditions

2.2.1.1. Maximal process volumes

Boiler:	10 L (2.64 Gal)
Distillate glass receiver:	0.6 L (0.16 Gal)
316L Feeding tank:	200 L (52.8 Gal)
316L Receiver tanks:	200 L (52.8 Gal)
316L Crash tank:	50 L (13.2 Gal)

2.2.1.2. Controlled parameters

- Column drop pressure DPT01
- Steam pressure PIC01
- Pre heating temperature TIC25
- Column feeding flow rate FIC01
- Re-boiler level LIC01
- Condenser level LIC02
- Reflux ratio: Reflux ratio controlled by a constant rate L/D by the ratio "FIC02/FT04"
- Vacuum pressure PIC02
- Cooling water flow rate FIC06

2.2.1.3. Working Conditions

The controlled distillation unit has been designed to give good experimental results under the following process conditions:

Process:	continuous
Feeding flow rate:	0 to 100 L/h (0 to 26.4 Gal/h)
Weight composition of the mixture:	0.30 wt. % (Monoethylene Glycol/water)
Operating pressure:	100 mbar (1.45 psi) to Atmospheric pressure
Distillate global flow rate:	0 to 80 L/h of distillate (0 to 11.0 Gal/h)
Reflux ratio:	0 to ∞ , nominal value of 0.3 to 0.5
Differential pressure in the column:	10 mbar to 18 mbar (0.15-0.26 psi)
Condenser water flow rate:	0.2 to 50 L/min, flow rate control (78 Gal/h)
Pre heating temperature:	43.33°C (110°F) for feeding on the 11 th tray

2.2.2. Weight and dimensions of the unit

Dimensions: W x D x H = 555 x 545 x 425 cm (219 x 215 x 167 in)
 Net weight: 4 250 kg (9 370 lbs)

2.2.3. List of chemical products used with the unit

Chemical products used on the unit must be compatible with the materials constituting the unit which are listed below:

- Demineralised Water
- Monoethylene Glycol N° CAS: 107-21-1

It is the responsibility of the operator to provide near the unit the list and MSDS sheets for the safe use of these products.

Remind:

It is indispensable to empty and clean the pilot as part of process cleanliness but also for the safety of the operator and to limit the risk of corrosion of the material making up the unit.

WARNING

**Bleach and any other chlorinated products are forbidden on this unit.
If left several days in the unit they can permanently damage the equipment.**

2.2.4. Materials

The unit is designed to be compatible with a large number of chemical products and especially with those given below.

The materials for the parts in contact with the reactants and synthesized products are:

- Glass
- Polytetrafluoroethylene (PTFE)
- 316 L Stainless Steel
- Polyetheretherketone (PEEK)
- Viton (FKM)

Note:

Always take into account the concentrations and process parameters to make sure that materials are compatible with the products you use.

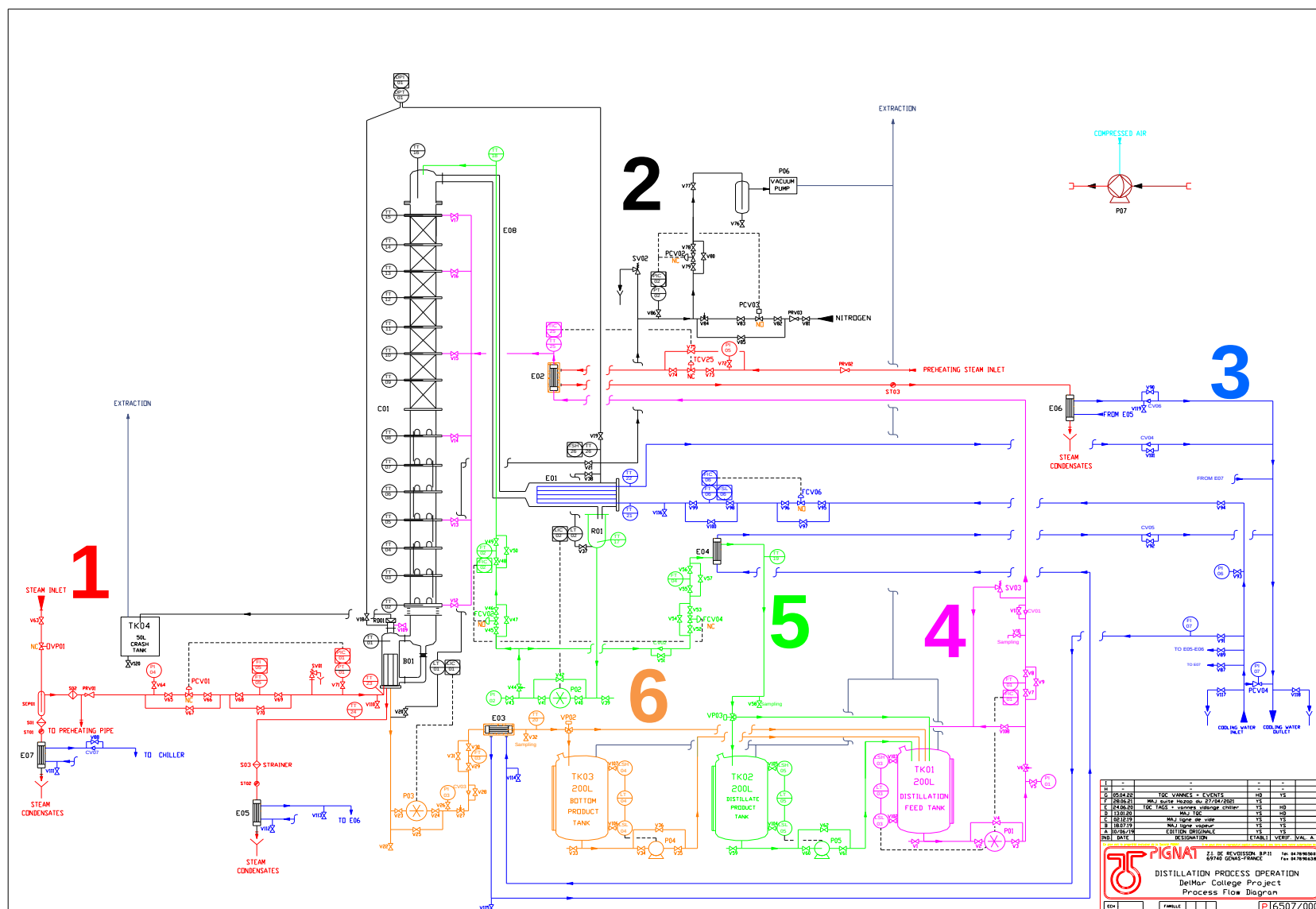
2.2.5. Functioning limits

The unit is **NOT** designed to:

- Be used with products likely to cause clogging
- Be used in working conditions outside those specified in this manual or on the unit label
- Be used by someone smoking
- Be used near a naked flame or source of sparks
- Be operated without surveillance or with operators who have not been taught how to use the unit correctly
- Be used with products which are incompatible with materials making up the unit

During maintenance operations, the replacement parts must conform to the original parts.

2.3. PRINCIPAL SCHEMATIC



1	Boiler Heating and Feeding pre-heating lines composed of: - Process ball valves to select either one or both steam lines - Safety valve SV01 to protect the whole steam lines - Pneumatic control valves TCV25 and PCV01 to adjust the pre-heating steam flow and the boiler steam pressure - Pressure regulator valves to adjust the steam pressure so that the temperature - Temperature transmitters for the steam - Pressure indicators for the steam - Flow and pressure transmitters for the boiler steam
2	Vacuum and nitrogen process lines composed of: - Vacuum pump P06 - Trap vessel to collect any traces of liquid before P06 - Pneumatic control valves PCV02/PCV03 working in opposite to adjust the system pressure Vacuum/Nitrogen - Pressure control loop PIC02 - Safety valve SV02 to prevent any overpressure - Rupture disk RD01 to prevent any overpressure up to 450 mbar (6.53 psi) - Crash tank TK04 to retain the product when the rupture disk breaks - A pipe on the head of the catch tank to extract gas - Pressure regulator valve PRV03 to adjust the nitrogen pressure
3	Cooling water lines composed of: - Overflow valve with a pressure indicator PI07 to adjust the cooling water outlet pressure - Flow control valve FCV06 to adjust the water flow inside the main shell/tube exchanger E01 - E01 inlet (TT21) and outlet (TT22) water temperature - Ball valve and float flow FI07 meter to adjust manually the water flow on other lines - Check valves to force water to flow in a single direction
4	Feeding product line composed of: - Stainless steel tank 200 L (52.8 Gal) - Gear pump P01 - The flow rate is measured and controlled with flow transmitter (FT01) by means of a speed controller for the pump (P01) - A pressure dial gauge PI01 to control the pressure at the outlet of the pump P01 by means of a needle valve V6 - A safety valve SV03 to prevent any overpressure from liquid expansion when the system is stopped - Tank vent pipe connect to extract gas

5	Distillate product line composed of: - Glass receiver R01 equipped with a level control device LIC02 to keep constant the distillate level - Gear pump P02 - Pressure dial gauge PI02 to control the pressure at the outlet of the pump P02 by means of a needle valve V44 - The flow rates are measured and controlled with flow control loop & transmitters FIC02/FT04 to measure the Reflux/Distillate by means of a speed controller for the pump (P02) and pneumatic flow control valves FCV02 (reflux) and FCV04 (distillate) working in opposite - Spirec heat exchanger E04 allows to cool down the distillate - Temperature transmitters TT18/TT19 to measure the reflux/distillate temperatures - A three-way ball valve VP03 to select either to send the distillate back to the feed tank TK01 or the distillate tank TK02 - A centrifugal pump P05 to transfer the distillate from TK02 back to TK01 - Tank vent pipe connect to extract gas
6	Bottom product line composed of: - Glass boiler B01 equipped with a level control device LIC01 to keep constant the boiler level - Gear pump P03 - Pressure dial gauge PI03 to control the pressure at the outlet of the pump P03 by means of a needle valve V27 - The flow rate is measured with a flow transmitter FT03 by means of a speed controller for the pump (P03) - Spirec heat exchanger E03 allows to cool down the bottom product - Temperature transmitters TT01/TT20 to measure the temperature in the boiler and out of E03 - A three-way ball valve VP02 to select either to send the bottom product back to the feed tank TK01 or the bottom tank TK03 - A centrifugal pump P04 to transfer the bottom product from TK03 back to TK01 - Tank vent pipe connect to extract gas

2.4. PARTS LIST

2.4.1. Components

TAG	DESCRIPTION	VOLUME / RANGE
TK01	316L stainless steel tank: for Feeding Product	200 L (52.8 Gal)
P01	316L / PEEK gear pump equipped with an internal safety valve to protect the pump	100 L/h (26.4 Gal/h) at 4 bars (58 psi)
CV01	316 stainless steel check valve	set at: 1.72 bar (25 psi)
TK02	316L stainless steel tank: for Distillate Product	200 L (52.8 Gal)
R01	Glass distillate receiver	0.6 L (0.16 Gal)
P02	316L / PEEK gear pump equipped with an internal safety valve to protect the distillate pump	80 L/h (21 Gal/h) at 4 bars (58 psi)
P05	316L stainless steel centrifugal pump to transfer the product back to the Feed Tank	2.4 m³/h (634 Gal/h) at 1.88 bars (27 psi)
CV02	316 stainless steel check valve	set at: 0.69 bar (10 psi)
E01	316L Mo/Glass Tubular heat exchanger: for Distillate Product	S=1.8 m² (19.38 ft²)
E04	316L SS Spirec Heat exchanger to cool down the distillate product	S=0.2 m² (2.15 ft²)
TK03	316L stainless steel tank to store the bottom product	0.2 m³ (52.8 Gal)
B01	Boiler with vertical tubular steam heat exchanger	S=0.7 m² (7.53 ft²)
P03	316L / PEEK gear pump equipped with an internal safety valve to protect the distillate pump	100 L/h (26.4 Gal/h) at 4 bars (58 psi)
P04	316L stainless steel centrifugal pump to transfer the bottom product back to the Feed Tank	2.4 m³/h (634 Gal/h) at 1.88 bars (27 psi)
CV03	316 stainless steel check valve	set at: 0.69 bar (10 psi)
E03	316L SS Spirec heat exchanger to cool down the bottom product from B01	S=0.2 m² (2.15 ft²)
SEP01	Carbon steel baffle type separator used for the removal of entrained liquids in steam	-
ST01	SG iron steam inlet ball float trap	-
ST02	SG iron boiler steam ball float trap	-

TAG	DESCRIPTION	VOLUME / RANGE
ST03	Stainless steel (with ENP) thermodynamic steam trap cycles periodically to discharge condensate very near to steam temperature	-
S01	SG iron Y-type strainer	-
S02	SG iron Y-type strainer	-
S03	SG iron Y-type strainer	-
E02	316L SS Spirec Heat exchanger to preheat the feed product	S=0.1 m ² (1.08 ft ²)
E05	316L SS Spirec Heat exchanger to cool down the steam from B01	S=0.1 m ² (1.08 ft ²)
E06	316L SS Spirec Heat exchanger to cool down the steam from E02	S=0.1 m ² (1.08 ft ²)
E07	316L SS Spirec Heat exchanger to cool down the steam SEP01	S=0.1 m ² (1.08 ft ²)
CV04	3 Pieces stainless steel EN 1.4408 check valve	-
CV05	3 Pieces stainless steel EN 1.4408 check valve	-
CV06	3 Pieces stainless steel EN 1.4408 check valve	-
CV07	3 Pieces stainless steel EN 1.4408 check valve	-
TK04	316L stainless steel tank to collect the liquid/gas product which come from RD01	50 L (13.2 Gal)
P06	PTFE sandwich diaphragm Vacuum Pump	3.4 m ³ /h (898 Gal/h) Ultimate vacuum: 1.5 mbar abs. (22 psi abs.)

2.4.2. Instrumentation

2.4.2.1. Flow transmitters

TAG	DESCRIPTION	RANGE	PILOT CONTROL
FT01	Variable area Flow transmitter for the Feed	9.8-98 L/h (2.59-25.89 Gal/h)	FIC01 Control
FT02	Variable area Flow transmitter for the Reflux	4.2-42 L/h (1.11-11.10 Gal/h)	FIC02 Control
FT03	Variable area Flow transmitter for the Bottom Product	9.8-98 L/h (2.59-25.89 Gal/h)	-
FT04	Variable area Flow transmitter for the Distillate	4.2-42 L/h (1.11-11.10 Gal/h)	-
FT05	Vortex Flow transmitter for the Heating Steam Inlet	0-75 kg/h (0-165 lbs/h)	-
FT06	Electromagnetic Flow transmitter for the Water through E01	0.2-50 L/min (3.17-792.52 Gal/h)	FIC06 Control

2.4.2.2. Flow meters

TAG	DESCRIPTION	RANGE	PILOT CONTROL
FI07	Float Flow meter for water through E03	100-1 000 L/h (26.42-264.17 Gal/h)	-

2.4.2.3. Pressure transmitters

TAG	DESCRIPTION	RANGE	PILOT CONTROL
PT01	Pressure Transmitter for the steam	0-10 bars rel. (0-145 psi rel.)	PIC01 Control
PT02	Pressure Transmitter for Vacuum/Nitrogen	0-1200 mbar abs. (0-17 psi abs.)	PIC02 Control

2.4.2.4. Pressure gauges

TAG	DESCRIPTION	RANGE
PI01	Pressure Gauge next to the Feed pump P01	0-6 bars (0-87 psi)
PI02	Pressure Gauge next to the Distillate pump P02	0-6 bars (0-87 psi)
PI03	Pressure Gauge next to the Bottom pump P03	0-6 bars (0-87 psi)
PI04	Pressure Gauge for the steam Boiler	0-10 bars (0-145 psi)
PI05	Pressure Gauge for the Steam preheating	0-6 bars (0-87 psi)
PI06	Pressure Gauge for the water inlet	0-6 bars (0-87 psi)
PI07	Pressure Gauge for the water overflow valve	1-5 bars (14.5-72.5 psi)

2.4.2.5. DeltaP transmitter

TAG	DESCRIPTION	RANGE	PILOT CONTROL
DPT01	Column deltaP transmitter	0-100 mbar (0-1.5 psi)	-

2.4.2.6. Temperature transmitters

TAG	DESCRIPTION	RANGE	PILOT CONTROL
TT01	Boiler Temperature	-50 / 200°C (-58 / 392°F)	-
TT02	Temperature on column bottom (tray n°1)	-50 / 200°C (-58 / 392°F)	-
TT03	Temperature on column bottom (tray n°2)	-50 / 200°C (-58 / 392°F)	-
TT04	Temperature on column bottom (tray n°3)	-50 / 200°C (-58 / 392°F)	-
TT05	Temperature on column bottom (tray n°4)	-50 / 200°C (-58 / 392°F)	-
TT06	Temperature on column bottom (tray n°5)	-50 / 200°C (-58 / 392°F)	-
TT07	Temperature Transmitter on column bottom (tray n°6)	-50 / 200°C (-58 / 392°F)	-
TT08	Temperature Transmitter on column bottom (tray n°7)	-50 / 200°C (-58 / 392°F)	-
TT09	Temperature Transmitter on column bottom (tray n°8)	-50 / 200°C (-58 / 392°F)	-
TT10	Temperature Transmitter on column top (tray n°9)	-50 / 200°C (-58 / 392°F)	-
TT11	Temperature Transmitter on column top (tray n°10)	-50 / 200°C (-58 / 392°F)	-
TT12	Temperature Transmitter on column top (tray n°11)	-50 / 200°C (-58 / 392°F)	-
TT12	Temperature Transmitter on column top (tray n°12)	-50 / 200°C (-58 / 392°F)	-
TT13	Temperature Transmitter on column top (tray n°13)	-50 / 200°C (-58 / 392°F)	-

TAG	DESCRIPTION	RANGE	PILOT CONTROL
TT14	Temperature Transmitter on column top (tray n°14)	-50 / 200°C (-58 / 392°F)	-
TT15	Temperature Transmitter on column top (tray n°15)	-50 / 200°C (-58 / 392°F)	-
TT16	Column Head Temperature Transmitter	-50 / 200°C (-58 / 392°F)	-
TT17	Distillate temperature Transmitter	-50 / 200°C (-58 / 392°F)	-
TT18	Reflux temperature Transmitter	-50 / 200°C (-58 / 392°F)	-
TT19	Distillate Temperature Transmitter sent to TK02	-50 / 200°C (-58 / 392°F)	-
TT20	Bottom Temperature Transmitter sent to TK03	-50 / 200°C (-58 / 392°F)	-
TT21	Water E01 Inlet Temperature Transmitter	-50 / 200°C (-58 / 392°F)	-
TT22	Water E01 Outlet Temperature Transmitter	-50 / 200°C (-58 / 392°F)	-
TT23	Steam B01 Inlet Temperature Transmitter	-50 / 200°C (-58 / 392°F)	-
TT24	Steam B01 Outlet Temperature Transmitter	-50 / 200°C (-58 / 392°F)	-
TT25	Feed Preheating Temperature Transmitter	-50 / 200°C (-58 / 392°F)	TIC25 Control
TT26	Condenser E01 Vent Temperature (TSH16)	-50 / 200°C (-58 / 392°F)	-

2.4.2.7. Level transmitters

TAG	DESCRIPTION	RANGE	PILOT CONTROL
LT01	Boiler level transmitter deltaP measurement	0-100 mbar (0-1.5 psi) 0-100%	LIC01 Control
LT02	Distillate level transmitter deltaP measurement	0-100 mbar (0-1.5 psi) 0-100%	LIC02 Control
LT03	Feed tank level transmitter deltaP measurement	0-100 mbar (0-1.5 psi) 0-100%	-
LT04	Bottom product tank level transmitter deltaP measurement	0-100 mbar (0-1.5 psi) 0-100%	
LT05	Distillate product tank level transmitter deltaP measurement	0-100 mbar (0-1.5 psi) 0-100%	

2.4.3. Valves

2.4.3.1. Manual valves

TAG	DESCRIPTION	TAG	DESCRIPTION
V01	TK01 draining ball valve	V02	P01 inlet ball valve
V03	P01 outlet ball valve	V04	P01 by-pass ball valve
V05	PI01 isolating ball valve	V06	P01 adjusting needle valve
V07	FT01 inlet ball valve	V08	FT01 outlet ball valve
V09	FT01 by-pass ball valve	V10	Feed product sampling ball valve
V11	CV01 by-pass ball valve	V12	Column bottom tray N°1 ball valve
V13	Column bottom tray N°4 ball valve	V14	Column bottom tray N°7 ball valve
V15	Column top tray N°10 ball valve	V16	Column top tray N°13 ball valve
V17	Column top tray N°15 ball valve	V18	DPT01 isolating ball valve (B01 side)
V19	DPT01 isolating ball valve (E01 side)	V20	LT01 isolating ball valve
V21	LT01 isolating ball valve	V22	B01 draining ball valve
V23	P03 inlet ball valve	V24	P03 outlet ball valve
V25	P03 by-pass ball valve	V26	PI03 isolating ball valve

TAG	DESCRIPTION	TAG	DESCRIPTION
V27	P03 adjusting needle valve	V28	CV03 by-pass ball valve
V29	FT03 inlet ball valve	V30	FT03 outlet ball valve
V31	FT03 by-pass ball valve	V32	Bottom product sampling ball valve
V33	TK03 Draining ball valve	V34	P04 inlet ball valve
V35	P04 outlet ball valve	V36	P04 by-pass ball valve
V37	LT02 isolating ball valve	V38	LT02 isolating ball valve
V39	R01 draining ball valve	V40	P02 inlet ball valve
V41	P02 outlet ball valve	V42	P02 by-pass ball valve
V43	PI02 isolating ball valve	V44	P02 adjusting needle valve
V45	FCV02 inlet ball valve	V46	FCV02 outlet ball valve
V47	FCV02 by-pass ball valve	V48	FT02 inlet ball valve
V49	FT02 outlet ball valve	V50	FT02 by-pass ball valve
V51	CV02 by-pass ball valve	V52	FCV04 inlet ball valve
V53	FCV04 outlet ball valve	V54	FCV04 by-pass ball valve

TAG	DESCRIPTION	TAG	DESCRIPTION
V55	FT04 inlet ball valve	V56	FT04 outlet ball valve
V57	FT04 by-pass ball valve	V58	Distillate product sampling ball valve
V59	TK02 draining ball valve	V60	P05 inlet ball valve
V61	P05 outlet ball valve	V62	P05 by-pass ball valve
V63	Steam inlet gate valve	V64	PI04 isolating ball valve
V65	PCV01 inlet ball valve	V66	PCV01 outlet ball valve
V67	PCV01 by-pass ball valve	V68	FT05 inlet ball valve
V69	FT05 outlet ball valve	V70	FT05 by-pass ball valve
V71	PT01 isolating ball valve	V72	PI05 isolating ball valve
V73	TCV25 inlet ball valve	V74	TCV25 outlet ball valve
V75	TCV25 by-pass ball valve	V76	Vacuum trap receiver draining ball valve
V77	Vacuum isolating ball valve	V78	PCV02 outlet ball valve
V79	PCV02 inlet ball valve	V80	PCV02 by-pass ball valve
V81	Nitrogen feeding ball valve	V82	PCV03 inlet ball valve

TAG	DESCRIPTION	TAG	DESCRIPTION
V83	PCV03 outlet ball valve	V84	Nitrogen adjusting needle valve
V85	PCV03 by-pass ball valve	V86	PT02 isolating ball valve
V87	E07 inlet water ball valve	V88	CV07 by-pass ball valve
V89	E05-E06 inlet water ball valve	V90	CV06 by-pass valve
V91	E03 inlet water ball valve	V92	CV05 by-pass valve
V93	PI06 isolating ball valve	V94	E01 inlet water ball valve
V95	FCV06 inlet ball valve	V96	FCV06 outlet ball valve
V97	FCV06 by-pass ball valve	V98	FT06 inlet ball valve
V99	FT06 outlet ball valve	V100	FT06 by-pass ball valve
V101	CV04 by-pass ball valve	V102	LT03 isolating ball valve
V103	LT03 isolating ball valve	V104	LT05 isolating ball valve
V105	LT05 isolating ball valve	V106	LT04 isolating ball valve
V107	LT04 isolating ball valve	V108	TK01 by-pass ball valve
V109	Boiler filling ball valve	V110	Steam inlet draining ball valve

TAG	DESCRIPTION	TAG	DESCRIPTION
V111	E07 inlet water draining ball valve	V112	E05 inlet water draining ball valve
V113	E05 outlet water draining ball valve	V114	E03 inlet water draining ball valve
V115	E03 outlet water draining ball valve	V116	E01 inlet water draining ball valve
V117	Cooling water inlet draining valve	V118	Cooling water outlet draining valve
V119	E06 outlet water draining ball valve	V120	TK04 Draining ball valve

2.4.3.2. Pneumatic valves

TAG	DESCRIPTION
VP01	Pneumatic safety valve to stop the steam inlet flow
VP02	Pneumatic 3-way valve to select either TK01 or TK03
VP03	Pneumatic 3-way valve to select either TK01 or TK02

2.4.3.3. Control valves

TAG	DESCRIPTION
PCV01	Pressure control valve (Normally Closed) to adjust the steam inlet pressure PIC01
FCV02	Flow control valve (Normally Opened) to adjust the reflux flowrate FIC02
PCV02	Pressure control valve (Normally Closed) to adjust the process vacuum PIC02
PCV03	Pressure control valve (Normally Opened) to adjust the process nitrogen feeding PIC02
FCV04	Flow control valve (Normally Closed) to adjust the distillate flowrate FT04
PCV04	Pressure control valve to adjust the cooling water inlet pressure PI07
FCV06	Flow control valve (Normally Opened) to adjust the cold-water flowrate FIC06
TCV25	Temperature control valve (Normally Closed) to adjust the preheating temperature TIC25

2.4.3.4. Pressure reducing valves

TAG	DESCRIPTION	RANGE
PRV01	Pressure reducing valve to adjust the heating steam inlet pressure	3.5-8.6 bars (51-125 psi)
PRV02	Pressure reducing valve to adjust the preheating steam inlet pressure	1.4-4 bars (20-58 psi)
PRV03	Pressure reducing valve to adjust the nitrogen inlet pressure	0-140 mbar (0-2 psi)

2.4.4. Safety devices / Alarms

TAG	DESCRIPTION	TYPE	LIMIT
LSL01	Boiler B01 Low level safety alarm	Analog	45%
LSH01	Boiler B01 High level safety alarm	Analog	90%
LSL02	Glass receiver R01 Low level safety alarm	Analog	30%
LSH02	Glass receiver R01 High level safety alarm	Analog	50%
LSL03	Feed Tank TK01 Low level safety alarm	Analog	10%
LSH03	Feed Tank TK01 High level safety alarm	Analog	80%
LSL04	Bottom Tank TK03 Low level safety alarm	Analog	10%
LSH04	Bottom Tank TK03 High level safety alarm	Analog	80%
LSL05	Distillate Tank TK02 Low level safety alarm	Analog	10%
LSH05	Distillate Tank TK02 High level safety alarm	Analog	80%
PSL02	Process PT02 Low pressure safety alarm	Analog	50 mbar (0.73 psi)
PSH02	Process PT02 High pressure safety alarm	Analog	1050 mbar (15.23 psi)
TSH26	Vent TT26 High temperature safety alarm	TOR	40°C (104°F)
FSL06	Cold-water FT06 Low flowrate safety alarm	Analog	1.32 m³/h (350 Gal/h)
SV01	Steam safety valve	Mechanical	8.27 bars (120 psi)
SV02	Overpressure mechanical safety valve	Mechanical	-
SV03	Feed safety valve	Mechanical	6 bars (87 psi)
RD01	Boiler rupture disk	Mechanical	0.45 bar (6.53 psi)

2.5. CONTROL BOX

Control box including:

- **1** Main switch and white power on indicator light

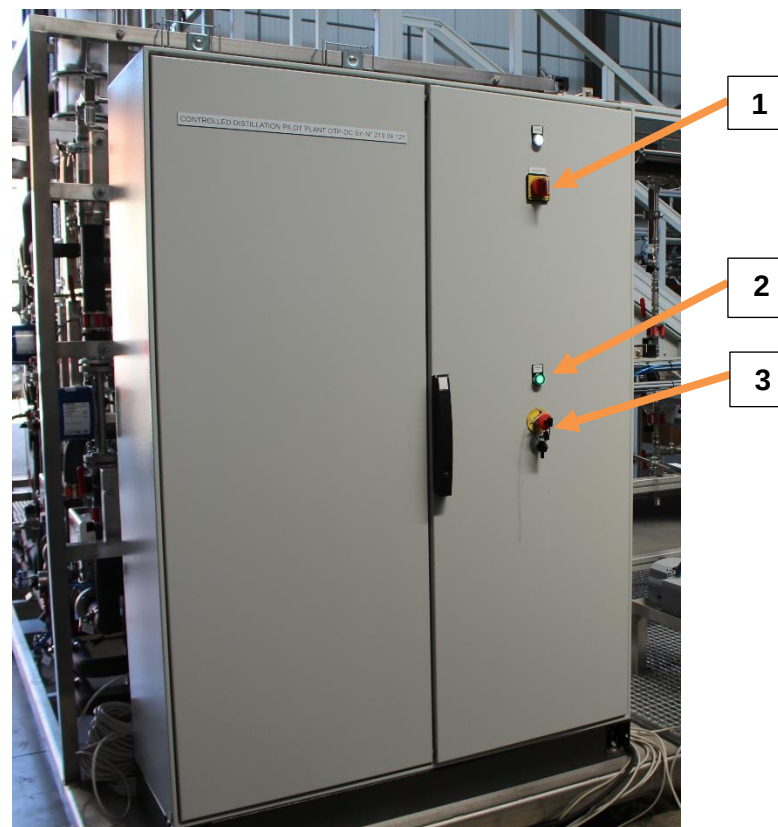
Activating the main switch, you will put the control box under voltage. The white light indicator shows that the electrical box is under voltage. But at this stage it is impossible to activate “power devices”, for example pumps or and control valves.

- **2** General push button to power up the instrumentation and the process control.

To power up the unit, press the power up general push button then its light becomes green.

- **3** Lockable emergency stop button type punch

When the emergency stop is pushed, power electrical devices are powered off. All controllers and indicators remain under voltage so that process parameters of the unit are displayed and the unit state is known.

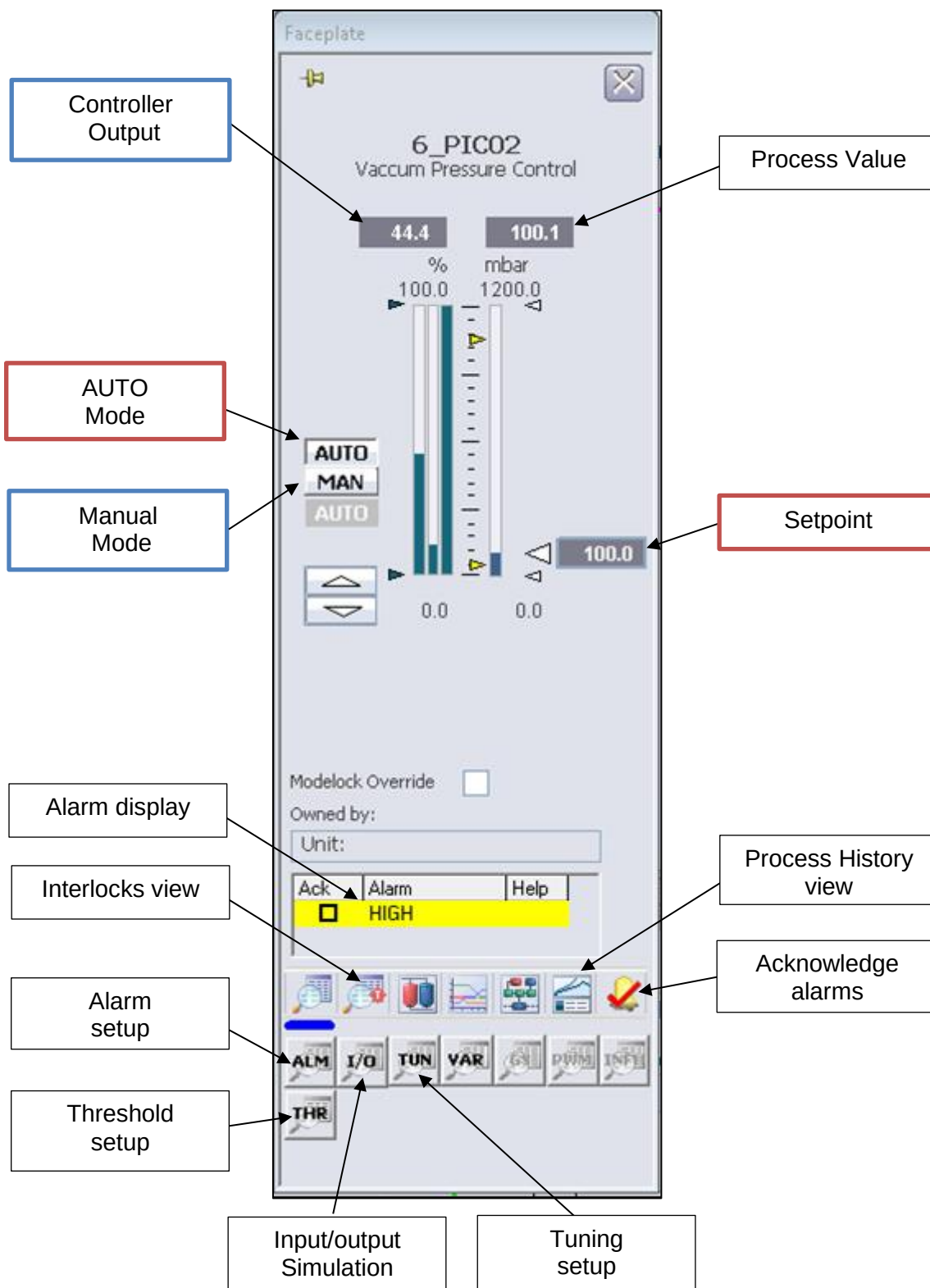


Electrical Plan N° 6349

2.6. CONTROLLER

2.6.1.1. Faceplate description

- Presentation of main control button on a DeltaV:



2.6.1.2. Faceplate details: Simulate I/O

- Used to simulate an analog Input for instance AI1
- Click on “Sim enable” box
- Then enter the parameter in “Sim Value (PV)” box

Detail

6_LIC01
Boiler Level Control
Dist Col

I/O

Simulate

	Field Value	Sim Enable	Sim Value(PV)	EU0	EU100	Units
AI						
AI1	86.0	☐	86.0	0.0	100.0	%
AI2	10.9	☐	10.9	0.0	100.0	%
PID						
PID1	--	☐	75.1	0.0	100.0	%
AO						
AO1	33.4	☐	33.4	0.0	100.0	no units

2.6.1.3. Faceplate details: Tuning parameters

- Used to adjust PID parameters
- Click on Gain, Reset, or Rate boxes
- Then adjust the parameter value

Detail

6_LIC01
Boiler Level Control
Dist Col

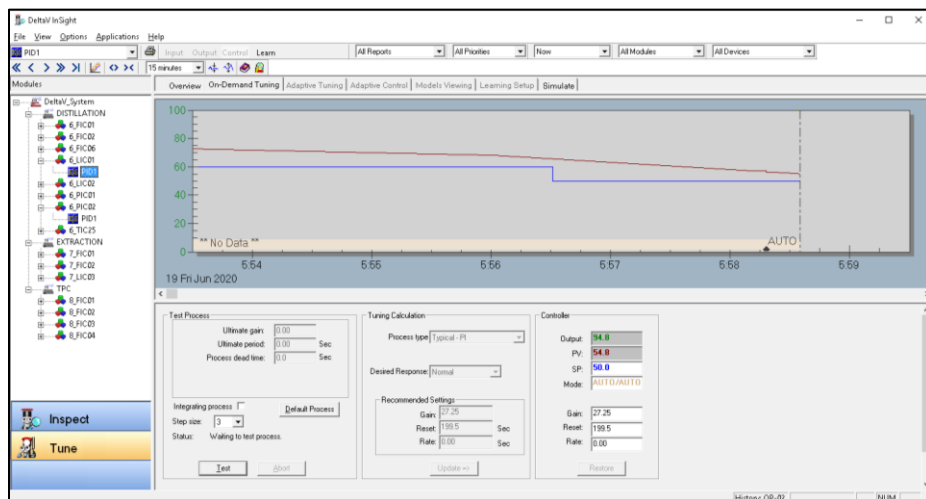
TUNING

Gain	27.25	Out Hi Lim	100.0		Restore ☐ Modelock
Reset	199.5 s	Out Lo Lim	0.0		
Rate	0.0 s	ARW Hi Lim	100.0		
PV Filter TC	0.0 s	ARW Lo Lim	0.0		
SP Filter TC	0.00 s	SP Hi Lim	100.0		
SP Rate	0.0 EU/s	SP Lo Lim	0.0		
SP Rate	0.0 EU/s	I Deadband	0.0		
Structure	PI action on error, D action on PV				
	AO1 Output Increase to Open				
PID Action	Direct				
PID Output	Increase to Open				

Insight

2.6.1.4. Faceplate details: DeltaV InSight

- Autotune function to find PID parameters
- Click on “*Insight*” button
- Then select one controlled parameter: LIC01, or LIC02...
- Select a calculation method: e.g. typical PI
- Click on test to run the calculation
- Last, you can update new PID parameters



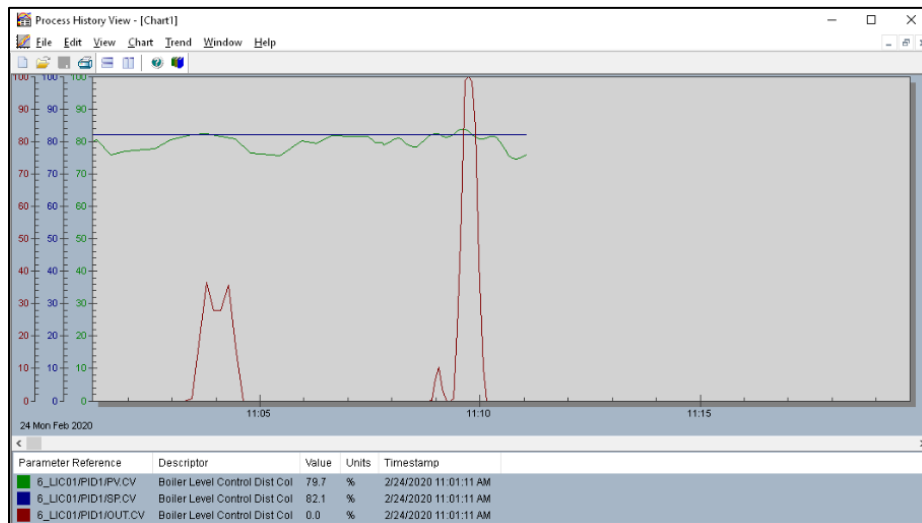
2.6.1.5. Faceplate details: Threshold

- Used to set up thresholds
- Tick off the “*Threshold Enable*” box
- Then enter the minimum value in “Lo” box, it means that if LIC01 < Threshold Lo, then the pump P03 which control the boiler level LIC01 will be put on standby.



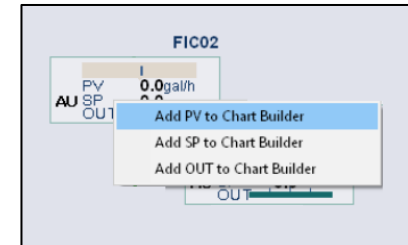
2.6.1.8. Faceplate details: Process History view

- Real-time display of the controlled process parameter for instance to control LIC01 level we have:
 - o LIC01/PID1/PV.CV = Process value
 - o LIC01/PID1/SP.CV = Setpoint
 - o LIC01/PID1/OUT.CV = Output (pump P03 actuator)
- Click on “*Process history view*” on the LIC01 faceplate to display the whole three parameters on a single chart.

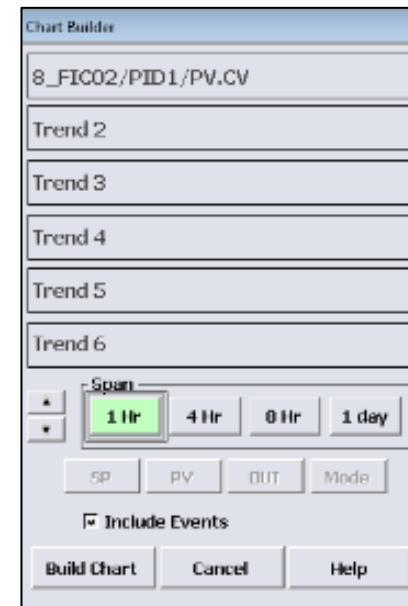


2.6.1.9. Build chart or add trend

- Right click on the faceplate
- Select faceplate's parameters you want to add to chart builder for instance PV, or SP...



- Then build the chart



- If you want to add new trends, right click on the chart
- Add trend to chart...
- Browse
- Select the area “*DISTILLATION*”
- Then select the parameters you want to display using the syntax: “*6_FIC02 > PID1 > PV > CV*”

2.7. CONTROLLERS PID CONFIGURATION:

	FIC01	FIC02	FIC06	LIC01	LIC02	PIC01	PIC02	TIC25
GAIN	1.04	0.2	1.18	27.25	8	4.45	20.5	0.95
RESET (S)	2.4	5.0	3.8	199.5	100	155	59.9	51.8
RATE (S)	0	0	0.6	0	0	0	0	8.3

2.8. SAFETY INTERLOCK FILE

CONDITIONS			Sensors - Description		Emergency Stop	FT08 (bath) Cooling water Flowrate	TT25 (VF) Preheating outlet	T H08 (VF) Vent Thermostat	DPT01 (kPa) Column differential pressure	PT01 (ps) Steam inlet pressure	P H02 (mbar) High vacuum pressure	P AL02 (mbar) Low vacuum pressure	LT01 (%) Reboiler level	LT02 (%) Reflux tank level	LT03 (%) Feed tank Level	LT04 (%) Bottom Product Tank Level	LT06 (%) Distillate Product Tank Level	DEFAULT Pump P01	DEFAULT Pump P02	DEFAULT Pump P03	DEFAULT Pump P04	DEFAULT Pump P05	DEFAULT Pump P06
ACTIONS																							
Type	Value																						
Signal	Unit																						
Block	Description	Actionneur	Action	NO (X)	TOR	ANA	ANA	TOR	ANA	ANA	ANA	ANA	ANA	ANA	ANA	ANA	TOR	TOR	TOR	TOR	TOR	TOR	
Steam	Boiler steam control valve PCV01 en MANU à 0%	PCV01 (INC)	Force OutPut 0%	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Steam	Preheating steam control valve TCV25 en MANU à 0%	TCV25 (INC)	Force OutPut 0%	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Distillate product	Distillate reflux control valve FCOV02 0% Closed 100% Open	FCOV02	Force OutPut 100 %	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Distillate product	Distillate reflux control valve FCOV04 0% Open 100% Closed	FCOV04 (INC)	Force OutPut 100 %	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Cooling water	Cooling water control valve FCOV06 en MANU à 0%	FCOV06 (INC)	Force OutPut 100 %	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Process Vacuum	Vacuum pressure control valve PCV02 (0-50%)	PCV02	Force OutPut 100 %	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Process Vacuum	Vacuum Vent control valve PCV03 (0-100%)	PCV03	Force OutPut 100 %	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Steam	Steam pneumatic safety valve	VP01 (INC)	STOP	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Bottom product	Pneumatic 3-ways valve VP02 to Feed or Bottom Prod Tank	VP02	Position "F" to Bottom Tank	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Distillate product	Pneumatic 3-ways valve VP03 to Feed or Distillate Prod Tank	VP03	Position "F" to Distillate Tank	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Feed product	Feed Pump P01 STOP	Pump P01	STOP	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Distillate product	Distillate Pump P02 STOP	Pump P02	STOP	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Bottom product	Bottom Pump P03 STOP	Pump P03	STOP	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Bottom product	Bottom Prod Tank Pump P04 STOP	Pump P04	STOP	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Distillate product	Distillate Prod Tank Pump P05 STOP	Pump P05	STOP	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Process Vacuum	Vacuum Pump P06 STOP	Pump P06	STOP	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Operator Alarms			Alarm	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	

2.1. MONITORING

2.1.1. Objectives

The software for controlled Vacuum Continuous Distillation unit has been developed to overview and control it from a computer.

The remote control gives the following possibilities:

- See indications of all analogical process parameters of the unit
- Enter the control loops set points
- Display of alarms
- Printing of alarm messages
- Overview historical data

Powering up the pump and the electrical actuator of the unit should be down from the software.

The didactical aims concerning the unit remote control software are:

- to follow and understand the evolution of the process parameters from synoptic windows, controller windows and historical graphs.
- to carry out a discontinuous distillation process using remote control.
- to study the experiments historical and organize their storage

Regulated parameters are:

- Feed flowrate FIC01
- Distillate flowrate FIC02
- Inlet cold water flowrate FIC06
- Reboiler level LIC01
- Distillate level LIC02
- Steam inlet pressure PIC01
- Column process pressure PIC02
- Feed preheating temperature TIC25

2.1.2. Changes

Any change or configuration modifications after the delivery of the unit and the training session by Pignat engineer are forbidden. Pignat responsibility will not be involved in case of incident or accident due to an inappropriate change of these factory configurations.

Changes of the overview screens can also be done in the presence of a computing Pignat technician.

2.1.3. Communication between regulators and monitoring

The rule of communication between the supervision system and the regulators is:

The last one who spoke is right

The unit is set from the different screens (synoptic and control) and from the controllers on the control box in accordance with the communication rule.

All settings are saved as soon as the IT supervision is started.

2.2. UTILITIES REQUIRED

2.2.1. Electricity supply

3x4000V + N + G – 50 Hz network
Power consumption: 12A

The electricity supply requires:

- A high sensitive differential protection of 30 mA.
- A connection (standard plug) which can be disconnected or Canalis®.

2.2.2. Cooling water supply

Cooling water enters in the unit by V117 (3/4"G) and outs by V118 (1/2"G).
Pressure: 3 bars (43.51 psi)
Flow rate: 5 m³/h (1 321 Gal/h)

2.2.3. Steam supply

Steam enters the unit by the V63 (DN15) gate valve.
Pressure: 8 bars (116.03 psi)
Flow rate: 48 kg/h (106 lbs/h)

2.2.4. Steam Condensate

A waste water discharge point is required continuously to drain the waste water from the steam condensate through a hose with an external diameter of less than 40 mm (1.57 in).

Recommendation: two separate points in Ø50 mm (Ø1.97 in) or a gutter.

2.2.5. Vacuum

The vacuum is generated by the vacuum pump.
Ultimate vacuum (abs): 1.5 mbar (0.0218 psi)
Max pumping speed at 50 Hz: 3.4 m³/h (898 Gal/h)
Power supply: via control cabinet

The technical documentation is provided in the technical file of the unit.

2.2.6. Compressed air

Compressed air is used to operate pneumatic and control valves.

Quality: deoiled and filtered dry air
Connection: on accessible client network
Pressure: 6 bars (87 psi) eff
Flow rate: 1 m³/h (264 Gal/h)

2.2.7. Nitrogen

Nitrogen is used to control the column's pressure.

Pressure $\leq$ 1 bar (14.5 psi)
Flow rate: 3 m³/h (793 Gal/h)

DISTILLATION PROCESS OPERATION

OTP-DC/ N° 22105073

OPERATING MANUAL

3. *SAFETY INFORMATION*

This manual contains important instructions which will enable the equipment to be used in a completely safe and appropriate way. Respect of these instructions will avoid risks, reduce the cost of repairs and the down time of the unit for these repairs, and finally, increase the reliability and life-time of the unit.

This manual must always be near the unit and available for consultation when the equipment is to be used.

This manual must be read by any person working with/on the unit, and the information applied.

In addition to this manual and to the regulations governing accident prevention and environmental protection in force in the country where the end user operates, recognized technical regulations have to be respected concerning safety and compliance with labor.

For any intervention on the unit, refer to this manual and the technical file.

This unit is intended for industrial use. It meets the standards and regulations in force. Specifications and approved conditions on site are included in this documentation. All instructions must be followed.

These instructions must be completed with the internal instructions of the operator.

3.1. RESPONSIBILITIES

3.1.1. Responsibility of the manufacturer

This unit has the same risks as those encountered on an industrial production unit. Its use must therefore be under the careful supervision of teachers/technicians instructed in its correct utilization, and in an appropriate place i.e. in accordance with the specifications given by the constructor.

If this unit is used in any way not specified in the technical recommendations given in this manual delivered with the unit, then PIGNAT cannot be held responsible for any use that does not comply with this instruction manual.

Any person working on this unit must be taught about its use and the risks encountered on this equipment.

3.1.2. Responsibility of the operator

Before carrying out any experiment on this pilot unit, the operator must investigate the compatibility of the products to be used and the material of the unit, and also the temperature and working pressure in the specified ranges given by the manufacturer.

Each person doing experiments on this unit must refer to the operating conditions given by the manufacturer in the technical file delivered with this unit.

The person in charge of the unit must also make sure that the correct safety measures are taken for working in a chemistry hall/lab i.e.:

- The wearing of appropriate personal protection equipment
- The correct transport of chemical products in the production area
- The correct storage of chemical products in an appropriate place
- The use of appropriate quantities with respect to the working volume of the unit
- Not using a cell phone close to the unit
- Not creating a naked flame close to the unit
- Not creating sparks close to the unit
- Etc.

This list is not exhaustive and must be completed by the person in charge of the unit.

3.2. RISKS INVOLVED

3.2.1. Chemical risk

3.2.1.1. Reminder of the products used with this unit

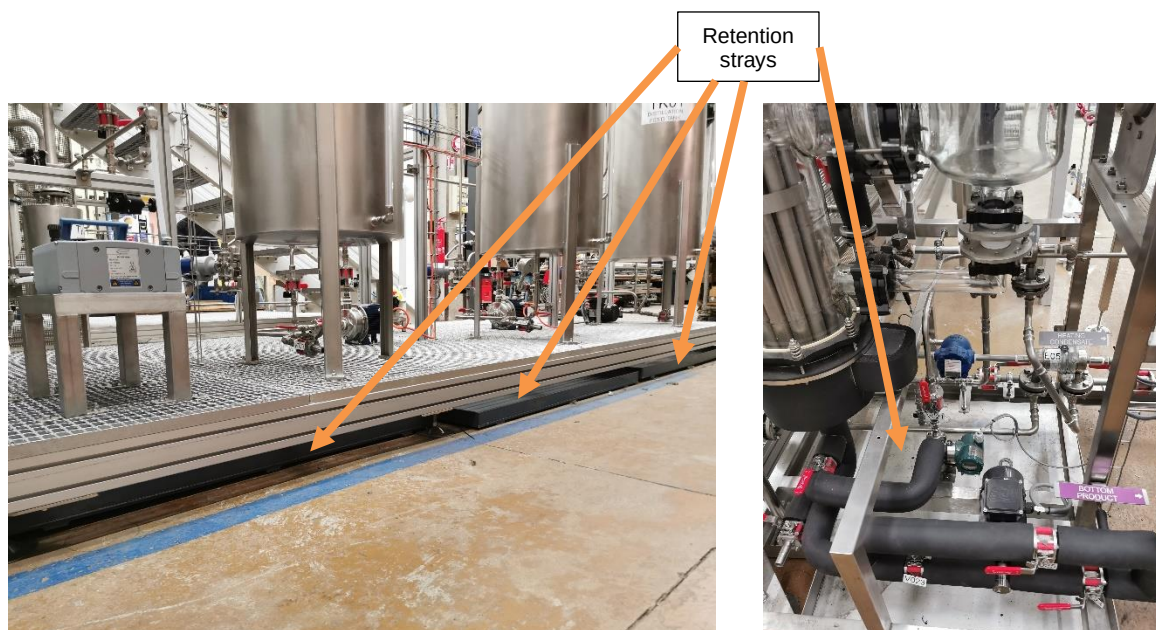
- Demineralised Water
- Monoethylene Glycol N° CAS: 107-21-1

On top of that it is possible to use cleaning or maintenance products. In this case the user will take all the required precautions.

These cleaning products must be compatible with the materials of the unit.

3.2.1.2. Risks associated with using the above products

The products used can present a significant risk for the operators handling them or working on this pilot unit (process operation and maintenance). It is therefore important for the operators to know the risks associated with their use. Four retention strays (145 cm x 120 cm x 5 cm / 57 in x 47 in x 2 in and 100 cm x 9 cm x 7 cm / 39 in x 4 in x 3 in for the boiler one) are also set up to retain the potential leaks during draining sessions.



We strongly recommend that all operators:

- **Read the material safety data sheet (MSDS) for each of the products**
- **Wear appropriate personal protection equipment (PPE) such as** cotton lab coat, safety glasses, face mask, gloves when handling these products
- **Respect all the instructions present in this operating manual**
- **Note down in the lab book all operations** realized on this unit.

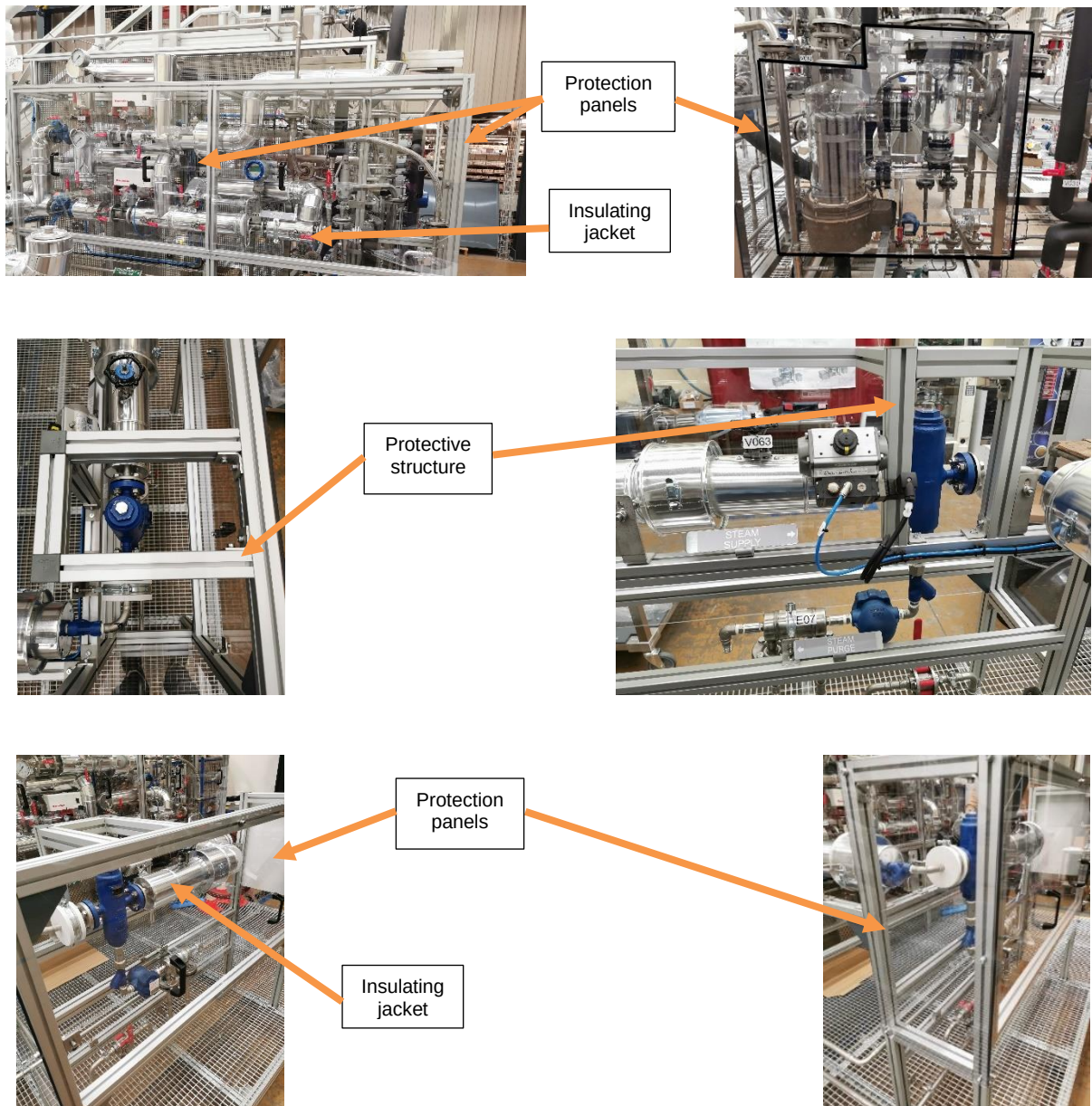
It is the responsibility of the operator to:

- Check the compatibility between the products used in the reactor and the materials in contact with these products.
- Make sure that the list of chemicals to be used and their MSDS are available close to the unit.

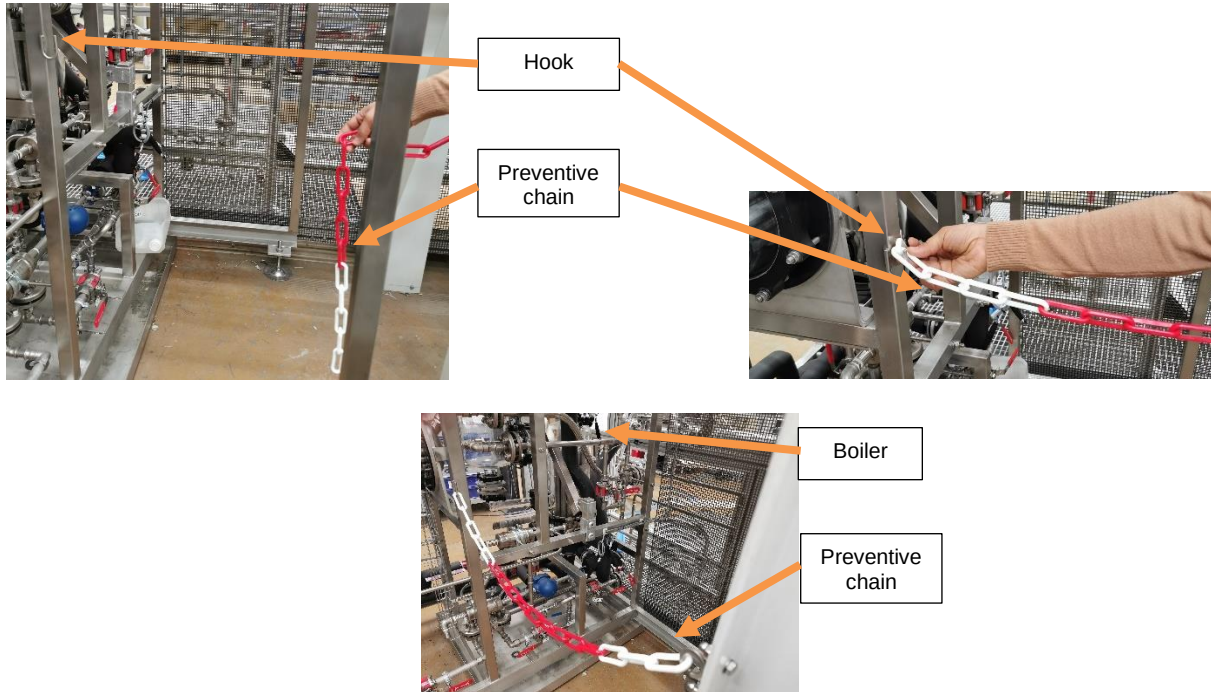
3.2.2. Thermal risk

Any part of the installation exceeding a surface temperature of 60°C (140°F) is considered hazardous. Prevention and / or protection means must be in place to limit the risk of burning. This unit uses steam to heat the boiler and the feed.

To protect users and reduce thermal losses, steam pipes are covered with an insulating jacket. On top of that there is a front protection panel on the reboiler and also on the steam lines, to prevent from burning. The separator SEP01 is equipped with a protective structure too, because it can't be covered with an insulating jacket.

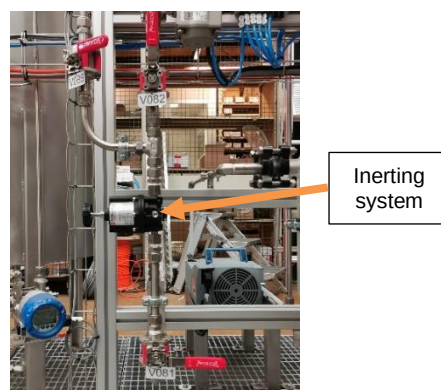


The user can only access to the boiler when he has to fill it. This filling can only be made before starting the heating. In other cases, he must stay behind the preventive chain. The user has to properly open and close the boiler's access by using the preventive chain and the hook:



A safety thermostat TSH26 set to **40°C (104°F)** in the control system cuts off the heating when the vent temperature from E01 reaches this value to avoid any risk of severe burning of users and the deterioration of some elements by heat. On top of that other safety analog parameters such as a low flowrate for cooling water in E01, or a high-pressure inlet steam, cuts off the heating by closing the inlet steam valve VP01.

The pressure regulator valve PRV03 allows to adjust the nitrogen pressure of the inerting system: it prevents oxygen from getting into the unit, which would lead to an ATEX zone. The regulation of the working pressure will be made by using the vacuum pump and nitrogen at atmospheric pressure, nitrogen being here an inert gas.



3.2.3. Electrical risk

Electrical risks are present during operation of the unit.

During maintenance operations:

- The persons working on the electrical elements must have the appropriate electrical qualifications to do so.
- The unit must be disconnected from all utilities (water, electricity) before any maintenance operation (electrical or mechanical).

3.2.4. Mechanical risks

- **Frame mobility**

If it is a fixed-type unit then there is no risk of movement. It is only necessary to verify that the unit is horizontal to avoid any risk of the unit tipping.

- **Rotating piece**

The moving parts of the pumps are not accessible so that there is no risk for the user.

- **How the different elements are held in place**

The design of the unit is such that all the elements are fixed on the chassis.

There is therefore no risk of any equipment falling off the chassis when the unit is in operation. When removing elements for cleaning or maintenance, the user must take the necessary precautions.

- **Movement of persons around and inside the unit**

To access the top part of the column there is a risk of falling or tripping from stair steps.

We put warning panels close to the main tripping risk, for instance the ladder, and also the raised chassis frame. It is up to the user to pay attention to this warning before getting around or inside the unit frame.



The user has also to respect the maximum load capacity (135 kg / 298 lbs) of the stairs written of the following panel.

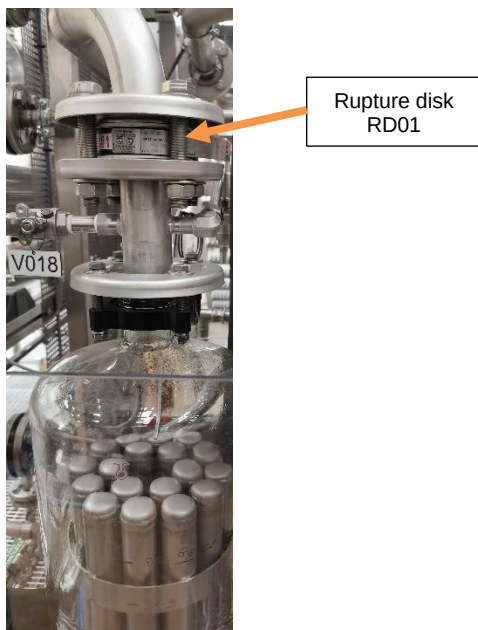


3.2.5. Risk of overpressure

Equipment may be subjected to PED if its operating pressure is strictly greater than +0.5 bar (7.25 psi) rel. The main risk remains is during the boiler heating and the vapours generated inside the glass column. That's why we put a safety valve SV01 calibrated at 8.27 bars (120 psi) on the steam inlet and a rupture disk RD01 calibrated at 0.45 bar (6.5 psi) to prevent from any overpressure on both sides' steam/process. If the rupture disk breaks, the catch tank will collect the gas/liquid product.



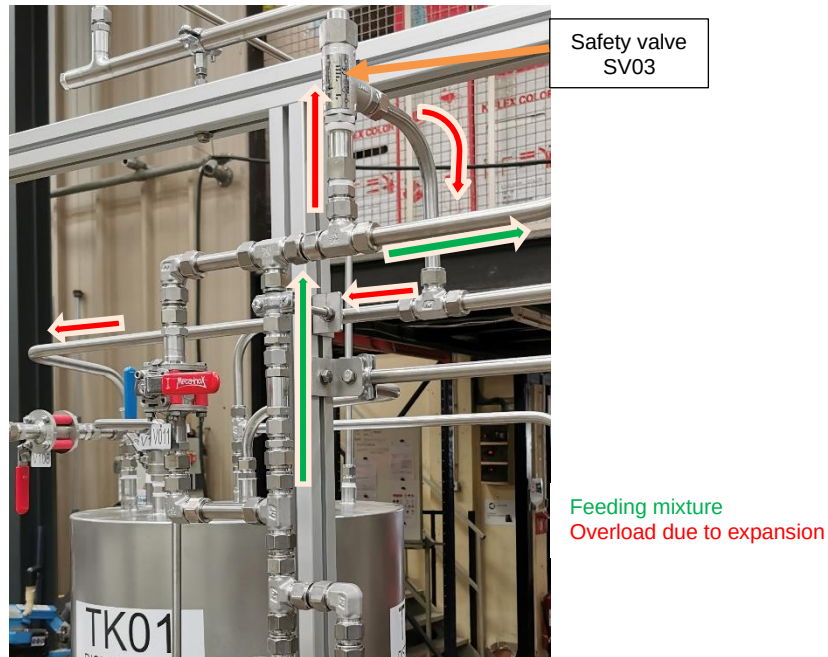
Extraction to the lab From the rupture disk RD01



In the case of this unit, the circulation of Water/MEG is due to the pumps. Their discharge pressure may be greater than 0.5 bar (7.25 psi) but less than 4 bars (58 psi) within the stainless-steel pipes and flexible hoses.

There is no risk during normal operation except for leaks in case of incorrect installation.
The circulating fluid is water/MEG at ambient to warm temperature so there is no major risk for the user.

When the circulations are stopped, some feeding mixture is blocked in the pipes between the E02 and CV01. If the user forgets to stop the heating on E02, the feeding mixture may expand within the pipes and cause an overpressure. This is prevented thanks to the safety valve SV03 by sending the overload back to the feed tank.



3.2.6. Noise risk

Noise is related to the pumps and the control valves (air exhaust) on this unit.

The sound level due to the equipment does not exceed 70 dB. However, the use of several machines in the same hall can make a great deal of noise. It is advisable to wear ear muffs or ear plugs.

3.3. ANY RESIDUAL RISK(S)

Because of the educational purpose of the unit, all risks cannot be reduced.
Nevertheless, the user must take into account the residual risks and remain attentive to the functioning of the unit and to his physical handling on it.

Risks that we have not been able to eliminate:

- Chemical risk

In all cases the users should handle with care when the unit is heated.
They will wear PPE which will protect them from residual risks.

DISTILLATION PROCESS OPERATION

OTP-DC/ N° 22105073

OPERATING MANUAL

4. *ASSEMBLY, INSTALLATION, OPERATION*

4.1. TRANSPORT / DELIVERY

Upon receipt of the equipment, check that it has not been damaged during transport.
In cases of damage through transport, make the usual reservations with the transporter.

Do not operate damaged equipment.

Use suitable and sufficiently strong handling equipment.

Remove the transport safety devices before commissioning.

4.2. INSTALLATION INSTRUCTIONS

4.2.1. Recommendations

The unit must be in a technical hall/lab adapted for the experiments/operations done with this unit.

We advise leaving space around the unit, about a 1 m (39.37 in) distance all-round the unit, in order to ensure easy access both for operation and maintenance.

These are the recommendations for the following operations:

- **Emptying/draining the unit:**
Emptying the unit is done by gravity, without any counter pressure of the liquid content, and must be done close to the pilot. Therefore, an evacuation system (drain) is necessary near the unit.
- **Lighting of the unit and work area:**
The unit should be used in an area with good and comfortable ambient lighting.
It is important to make sure that the unit is situated in a place with good lighting.
The addition of specific lighting will be realized by the client
- **Safety instructions:**
The safety instructions associated with the machine must be visible and close to the work unit, put there by the client.

4.2.2. Support frame of the unit

The design of the PIGNAT unit is free-standing i.e. all equipment is fixed on the steel frame.

4.2.3. Handling and fixing the unit

Permanently-positioned machines:

The handling of this type of machine is done by pallet truck.

The adjustable feet permit the unit to be levelled and to be fix to the ground.

4.2.4. Assembly

Factory assembly:

The machines are preassembled in the factory by technicians experienced in process engineering pilot equipment. This preassembly enables the realization of machine validation tests before the final installation with the client

Assembly on site:

The assembly on site is ensured by PIGNAT technicians. The assembly validation is done during the seal/watertight tests by the person responsible for the installation.

Warning

It is forbidden to touch an assembly when the unit is under vacuum or under pressure.

4.2.5. Positioning and setting up the unit

Warning

All interventions on PIGNAT units must be realized under the responsibility of a PIGNAT technician.

Permanently-positioned machines:

A pallet truck will be used to move and position the unit in the desired place.

4.3. CONNECTING TO THE UTILITIES

The connection to the utilities is realized by PIGNAT technicians, and the correct functioning is verified during the final validation tests on site.

The connection is made using fittings, gaskets and bolts adapted to operating conditions and properly sized.

Respect the direction of fitting, threads, seals...

Ensure that the equipment is free from stress.

The connection to the different networks is realized near the machine.

It is the operator responsibility to install on the utility networks the safety devices (safety valves, differential circuit-breaker...) required to work under the operating conditions specified by the manufacturer.

For this purpose, breakers, cut-off valves, regulators... must be installed on the utility networks by the operator on the connection with the unit to isolate the equipment and protect it against overpressure, overheating, overvoltage...

The machine is equipped with its own cut-off valves, which are easily accessible by the operator.

4.3.1. Electrical connections

The electrical connection is made on the 3x400V + N + G electrical network, required protection: 12 A.

The electricity supply requires:

- A high sensitive differential protection of 30 mA.
- A connection using a standard detachable plug or Canalis® .

Lay the power cord out of the passage area.

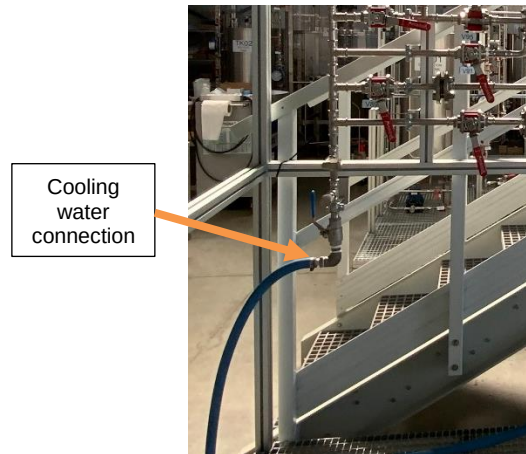
4.3.2. Connection to the Cooling water network

Cooling water feeds all heat exchangers in the unit.

Do not use straight deionized or reverse osmosis water in your cooling water supply.
The use of untreated or improperly treated water may result in scaling, or algae.

Cooling water plumbing connections to the unit:

- Tricoclair hose Ø25x33mm (Ø0.98x1.30in)

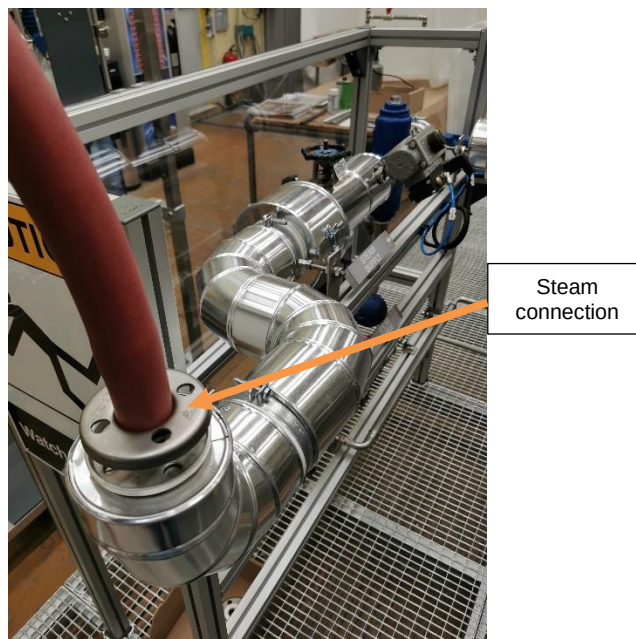


4.3.3. Connection to the steam

There is a single steam connection, for both boiler B01 and preheating heat exchanger E02.
Each circuit has its own valves set, pressure regulator and indicators.
The steam is connected before the general steam gate valve V63.

Steam plumbing connection:

- Stainless steel hose DN15

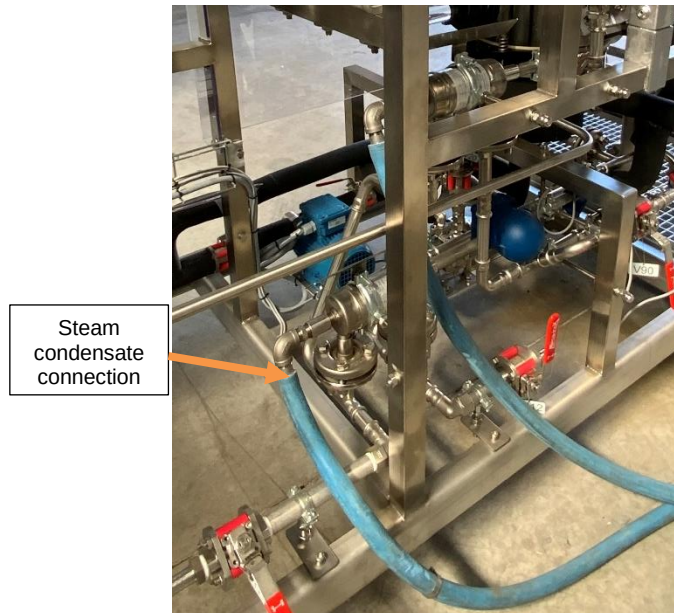


4.3.4. Connection to the evacuation of waste water

In this process, steam is cooled down through heat exchangers before getting evacuated.
Waste water comes from the steam condensates

Steam condensates evacuation plumbing connections:

- EPDM hose Ø19x30mm (Ø0.75x1.18in)



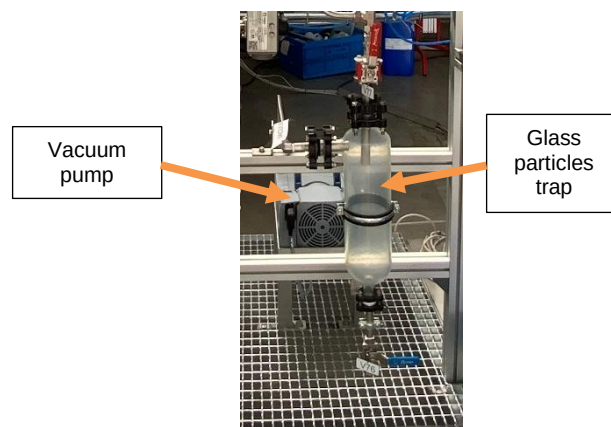
4.3.5. Vacuum connection

The vacuum pump is directly connected to a liquid trap glass, then the vacuum suction head is on the condenser E01.

The vacuum pump is integrated in the frame of the unit, so the connection is already done.

Vacuum pump plumbing connection:

- Outlet Hose nozzle DN 8-10 mm
- Inlet Hose nozzle DN 8-10 mm



4.3.6. Connection to the compressed air network

The compressed air is used to operate pneumatic and control valves.

AC coming from the compressor (not provided) is connected to the unit at a single point through a pressure reducing device. At this stage supply pressure must be set at 4 bars (58 psi). Then AC is distributed to the whole pneumatic equipment, control valves have their own pressure reducing device set at 2 bars (29 psi).

AC plumbing connection:

- PU hose Ø6x8mm (Ø0.24x0.31in)

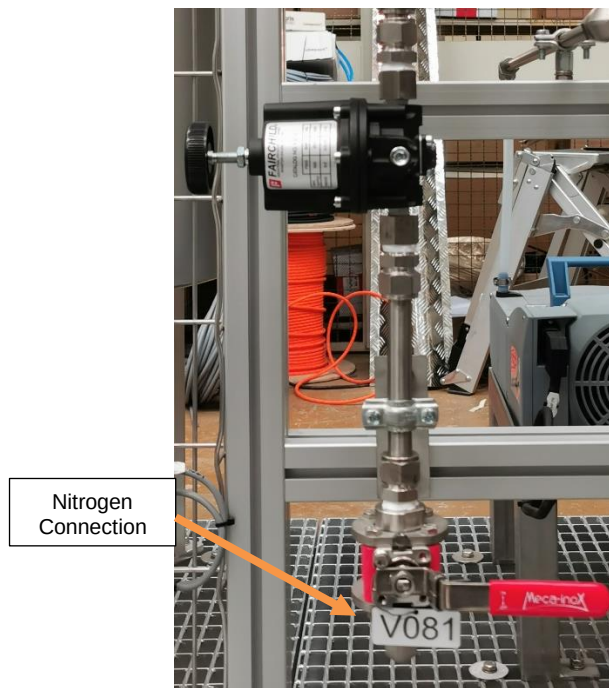


4.3.8. Connection to the nitrogen feeding

The process nitrogen has a single-entry point on V81.

Nitrogen plumbing connection

- PU hose Ø6x8mm (Ø0.24x0.31in)

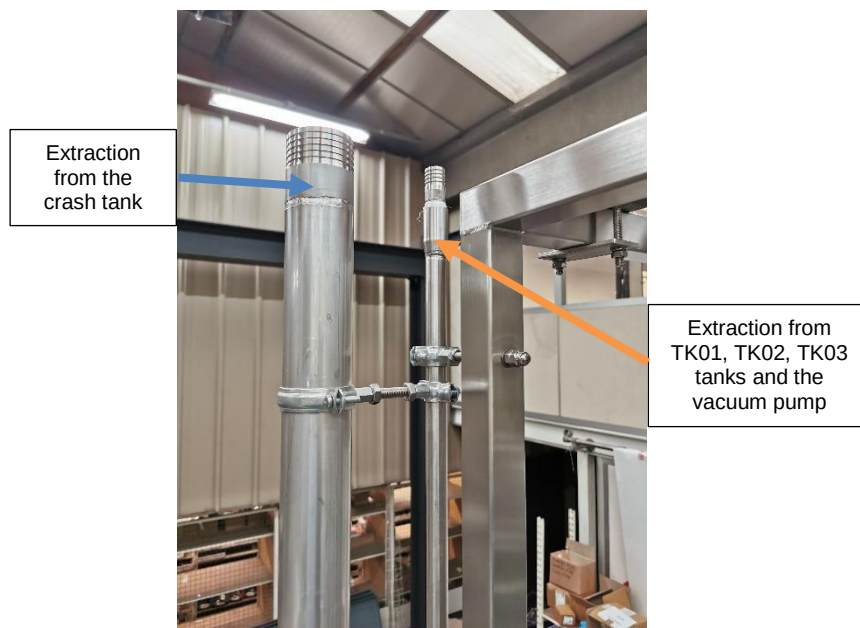


4.3.9. Connection to the extraction

The lab must extract gas from the four tanks in order to not let it stay in ambient air. The vacuum pump's exhaust must be extracted too. The extraction from the crash tank is set apart from the others ones.

Extraction plumbing connection:

- DN50 (for the crash tank)
- DN15 (for the other ones)



DISTILLATION PROCESS OPERATION

OTP-DC/ N° 22105073

OPERATING MANUAL

5. *OPERATING INSTRUCTIONS*

5.1. CHECKING THE UNIT

5.1.1. Verification of utilities

Check that the connections of the unit to the utilities are made.

- Electricity supply
- Steam supply
- Cooling water supply
- Waste water/steam condensate drainage system
- Compressed air supply
- Vacuum connection
- Nitrogen feeding

Connexion to utilities will be realized by Pignat technicians that will also check that everything is working correctly during the final tests in the customer's laboratory.

All utilities network must be accessible near the unit (maximal distance of 2 meters / 78.74 inches). If the customer wants, he can install additional isolating valves on each network before the unit.

Anyway, the unit always has its own isolation valves for fluids feedings. These valves are very easy to find on the unit and easy to access.

Power supply connexion: Connexion to the power supply network is done via a distribution box or a conduit box. The power network must be equipped with an upstream differential protection with high sensitivity of 30mA.

5.1.2. Checking the cleanliness of the unit

Read annotations recorded in the book of practical work to know the latest operations performed with the unit, to ensure that there are no indications against the operation you want to achieve.

Check the cleanliness of the unit:

NB: the cleaning of the unit is an integral part of the use of the unit and cleaning conditions have to be specified for each operation in the book of practical work. It is therefore important to check that the cleaning has been completed.

Check the following points:

- All the valves are closed
- On the external surfaces: no traces of reactants and products.
- On the internal surfaces: no traces of reactants and products.

If one of these points is not respected, it is important to realize the procedure of cleaning before using the unit again.

5.1.3. Status of the valves

All the valves of the installation are closed: only the valves required for the process will be opened.

The control valves are in their fallback position.

5.1.4. Installation guide lines

The unit must be installed in a laboratory adapted to chemical process units.

The following services are necessary:

Aeration of the room: The unit must then be installed in a room with adapted air refreshing to the use of chemical products. Use vapour mobile extractors near the points of charge and discharge of the products.

Working area lighting: The unit can be used with a standard lighting. Obviously, the lighting must be sufficient so that students see what they are doing. The customer can use additional lighting but this is at his charge.

Noise emission: The noise level created by the unit is not over 70 dB. The main sources of noise are the pumps.

Safety instructions: The general safety instructions should be shown next to the unit. This is charge of the educational team. Some symbolic stickers have been installed by Pignat on the unit to show the most hazardous areas on the unit.

5.2. POWERING THE UNIT

5.2.1. Utilities

Check that all utilities networks are working correctly and can be used: ask to the laboratory manager.

Check that networks isolation valves (on the networks) are opened:

- Compressed air
- Cooling water
- Nitrogen

Position correctly the evacuation hoses, in particular:

- Steam condensate hose
- Extraction hoses

5.2.2. Powering up the electric control box

Switch on the unit main switch and check that the white light indicator is active.

To power up the unit and allow all elements to work, press the power up general push button.

The light indicator of the general ON button is green.

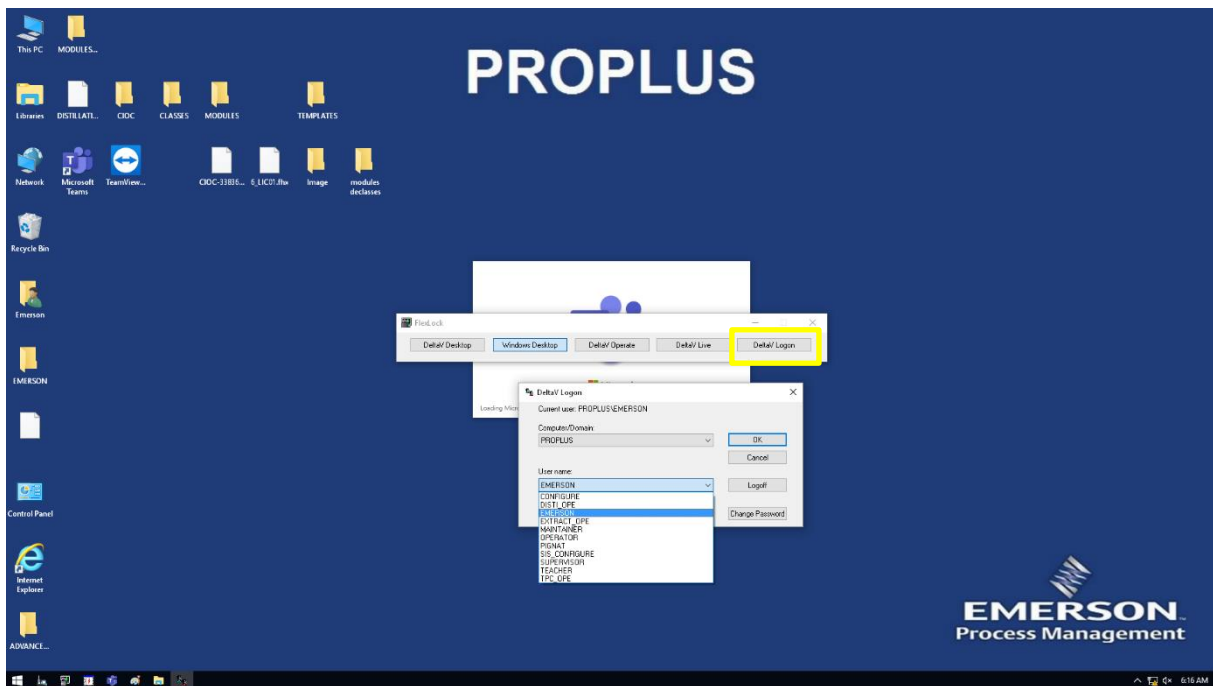
5.2.3. Starting the unit control software

Power up the computer.

Click on DeltaV Logon, then Log in under the appropriate username depending on your level access.

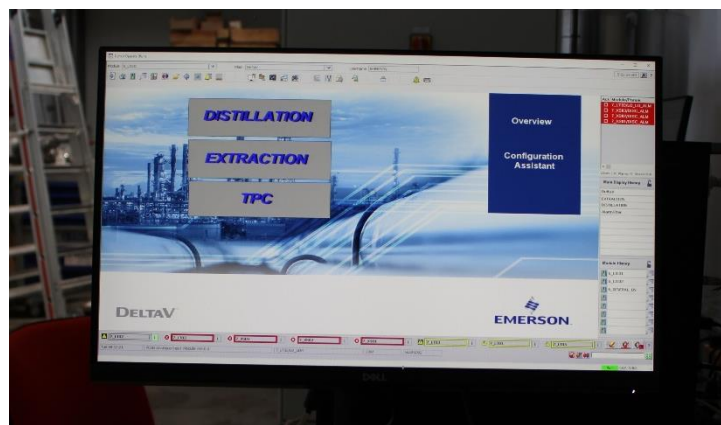
Username:

- EMERSON: higher access level, restricted for automation engineer
Password: **DeltaVE1**
- TEACHER: access level for teachers
Password: **teacher01**
- DISTI_OPE: access level for process operators
Password: **distillation**

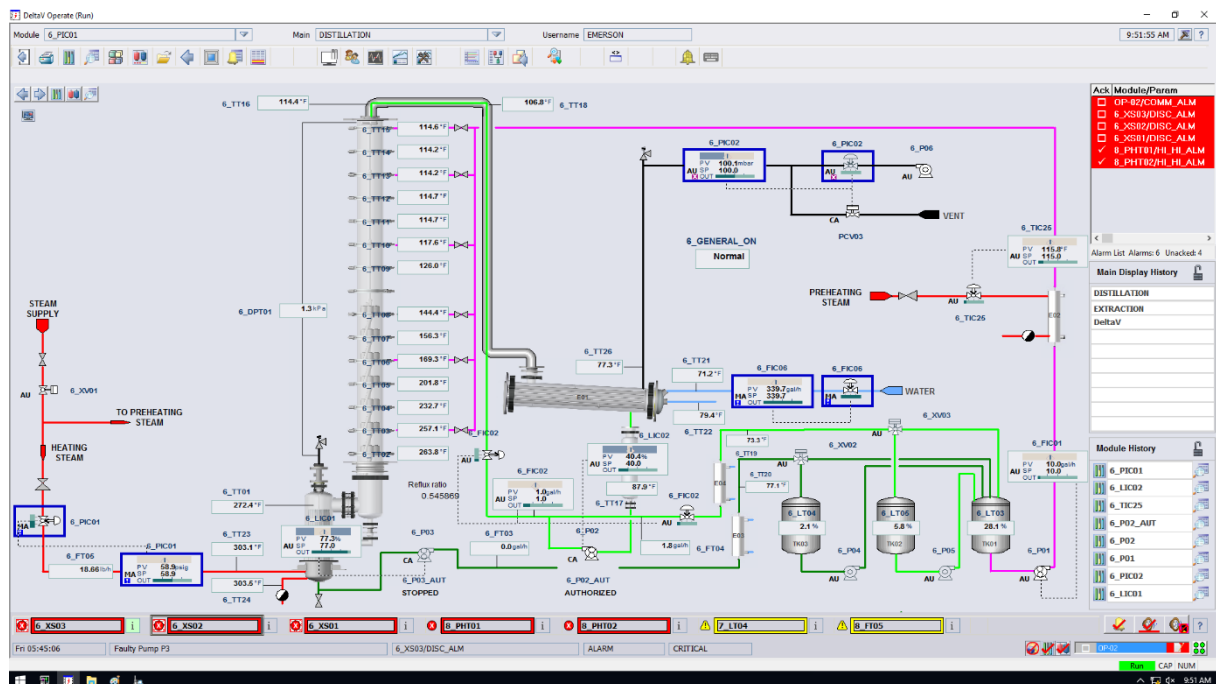


Login screen

Click on DeltaV Operate to run the application.



Select the area DISTILLATION.



Distillation running screen

5.2.4. Preparation of the unit before an experiment

Check that the unit is clean and empty. If necessary, refer to the unit laboratory book to know what was the last experiment on the Controlled distillation unit. Look at the products used and the distillation type carried out. In general:

- Open the feeding valves
- Open instrumentation valves
- Close the sampling valves
- Close the draining valves
- Close by-pass valves

Steam valves in normal running

TAG	Description	State
V63	Steam inlet gate valve	Closed
VP01	Pneumatic safety valve to stop the steam inlet flow	Closed
V64	PI04 isolating ball valve	Open
V65	PCV01 inlet ball valve	Open
V66	PCV01 outlet ball valve	Open
V67	PCV01 by-pass ball valve	Closed
V68	FT05 inlet ball valve	Open
V69	FT05 outlet ball valve	Open
V70	FT05 by-pass ball valve	Closed
V71	PT01 isolating ball valve	Open
V110	Steam inlet draining ball valve	Closed
V72	PI05 isolating ball valve	Open
V73	TCV25 inlet ball valve	Open
V74	TCV25 outlet ball valve	Open
V75	TCV25 by-pass ball valve	Closed

Cooling water valves in normal running

TAG	Description	State
V87	E07 inlet water ball valve	Open
V88	CV07 by-pass ball valve	Closed
V89	E05-E06 inlet water ball valve	Open
V90	CV06 by-pass valve	Closed
V91	E03 inlet water ball valve	Open
V92	CV05 by-pass valve	Closed
V93	PI06 isolating ball valve	Open
V94	E01 inlet water ball valve	Open
V95	FCV06 inlet ball valve	Open
V96	FCV06 outlet ball valve	Open
V97	FCV06 by-pass ball valve	Closed
V98	FT06 inlet ball valve	Open
V99	FT06 outlet ball valve	Open
V100	FT06 by-pass ball valve	Closed
V101	CV04 by-pass ball valve	Closed
V111	E07 outlet water draining ball valve	Closed
V112	E05 inlet water draining ball valve	Closed
V113	E05 outlet water draining ball valve	Closed
V114	E03 inlet water draining ball valve	Closed
V115	E03 outlet water draining ball valve	Closed
V116	E01 inlet water draining ball valve	Closed
V117	Cooling water inlet water draining ball valve	Closed
V118	Cooling water outlet water draining ball valve	Closed
V119	E06 outlet water draining ball valve	Closed

Feed product in normal running

TAG	Description	State
V1	TK01 Draining ball valve	Closed
V2	P01 inlet ball valve	Open
V3	P01 outlet ball valve	Open
V4	P01 by-pass ball valve	Closed
V5	PI01 isolating ball valve	Open
V6	P01 adjusting needle valve	½ Open
V7	FT01 inlet ball valve	Open
V8	FT01 outlet ball valve	Open
V9	FT01 by-pass ball valve	Closed
V10	Feed product sampling ball valve	Closed
V11	CV01 by-pass ball valve	Closed
V12	Column bottom tray N°1 ball valve	Closed
V13	Column bottom tray N°4 ball valve	Closed
V14	Column bottom tray N°7 ball valve	Closed
V15	Column top tray N°10 ball valve	Closed
V16	Column top tray N°13 ball valve	Open
V17	Column top tray N°15 ball valve	Closed
V102	LT03 isolating ball valve	Open
V103	LT03 isolating ball valve	Open
V108	TK01 by-pass ball valve	½ Open
V109	Boiler filling ball valve	Closed

Distillate product in normal running

TAG	Description	State
V37	LT02 isolating ball valve	Open
V38	LT02 isolating ball valve	Open
V39	R01 draining ball valve	Closed
V40	P02 inlet ball valve	Open
V41	P02 outlet ball valve	Open
V42	P02 by-pass ball valve	Closed
V43	PI02 isolating ball valve	Open
V44	P02 adjusting needle valve	½ Open
V45	FCV02 inlet ball valve	Open
V46	FCV02 outlet ball valve	Open
V47	FCV02 by-pass ball valve	Closed
V48	FT02 inlet ball valve	Open
V49	FT02 outlet ball valve	Open
V50	FT02 by-pass ball valve	Closed
V51	CV02 by-pass ball valve	Closed
V52	FCV04 inlet ball valve	Open
V53	FCV04 outlet ball valve	Open
V54	FCV04 by-pass ball valve	Closed
V55	FT04 inlet ball valve	Open
V56	FT04 outlet ball valve	Open
V57	FT04 by-pass ball valve	Closed
V58	Distillate product sampling ball valve	Closed
V59	TK02 Draining ball valve	Closed
V60	P05 inlet ball valve	Open
V61	P05 outlet ball valve	Open
V62	P05 by-pass ball valve	Closed
V104	LT05 isolating ball valve	Open
V105	LT05 isolating ball valve	Open

Bottom product in normal running

TAG	Description	State
V20	LT01 isolating ball valve	Open
V21	LT01 isolating ball valve	Open
V22	B01 draining ball valve	Closed
V23	P03 inlet ball valve	Open
V24	P03 outlet ball valve	Open
V25	P03 by-pass ball valve	Closed
V26	PI03 isolating ball valve	Open
V27	P03 adjusting needle valve	½ Open
V28	CV03 by-pass ball valve	Closed
V29	FT03 inlet ball valve	Open
V30	FT03 outlet ball valve	Open
V31	FT03 by-pass ball valve	Closed
V32	Bottom product sampling ball valve	Closed
V33	TK03 Draining ball valve	Closed
V34	P04 inlet ball valve	Open
V35	P04 outlet ball valve	Open
V36	P04 by-pass ball valve	Closed
V106	LT04 isolating ball valve	Open
V107	LT04 isolating ball valve	Open

Nitrogen/vacuum & Crash tank in normal running

TAG	Description	State
V76	Vacuum trap receiver draining ball valve	Closed
V77	Vacuum isolating ball valve	Closed
V78	PCV02 outlet ball valve	Open
V79	PCV02 inlet ball valve	Open
V80	PCV02 by-pass ball valve	Closed
V81	Nitrogen feeding ball valve	Open
V82	PCV03 inlet ball valve	Open
V83	PCV03 outlet ball valve	Open
V84	Nitrogen adjusting needle valve	½ Closed
V85	PCV03 by-pass ball valve	Closed
V86	PT02 isolating ball valve	Open
V120	TK04 draining ball valve	Closed

5.3. LOADING

5.3.1. Information

Before introducing any kind of liquid into the unit, please make sure that:

- The volumes and compounds are compatible with the unit and its materials.
- You wear adapted clothes and individual protections.
- The boiler temperature is lower than 30°C (86°F).

5.3.2. Manual filling of the reboiler

Use plastic vessels to charge the products. During this step the column must be operated at atmospheric pressure

Procedure:

- Open V81 (nitrogen valve).
- Check that valves V18 and V22 (boiler draining valve) are closed.
- Connect the discharge hose from the pneumatic pump to V109.
- Put the suction hose from the pump into your pure glycol tank.
- Open V109.
- Then start the pneumatic pump, gradually increasing the CA pressure from 0 to 2 bars (0 to 29 psi).
- Fill up the boiler with more or less 17 L (4.49 Gal) of pure glycol, LIC01 between 65-75%.



5.3.3. Manual filling of the 200 L (52.8 Gal) feeding Tank TK01

Use plastic vessels to charge the products. The materials should be big enough to avoid then can pass through the charging hole.

Connect the transfer pump to the charging hole

Fill the feeding storage tank TK01 with 200 L (52.8 Gal) of an aqueous solution containing at least 30% by mass of pure glycol.



5.4. VACUUM CONTINUOUS DISTILLATION

We explain in the following paragraphs how to do each type of operation necessary to carry out a continuous distillation unit.

The operating procedures have the objective for you to learn how to start, follow and make your own conclusions of an experiment.

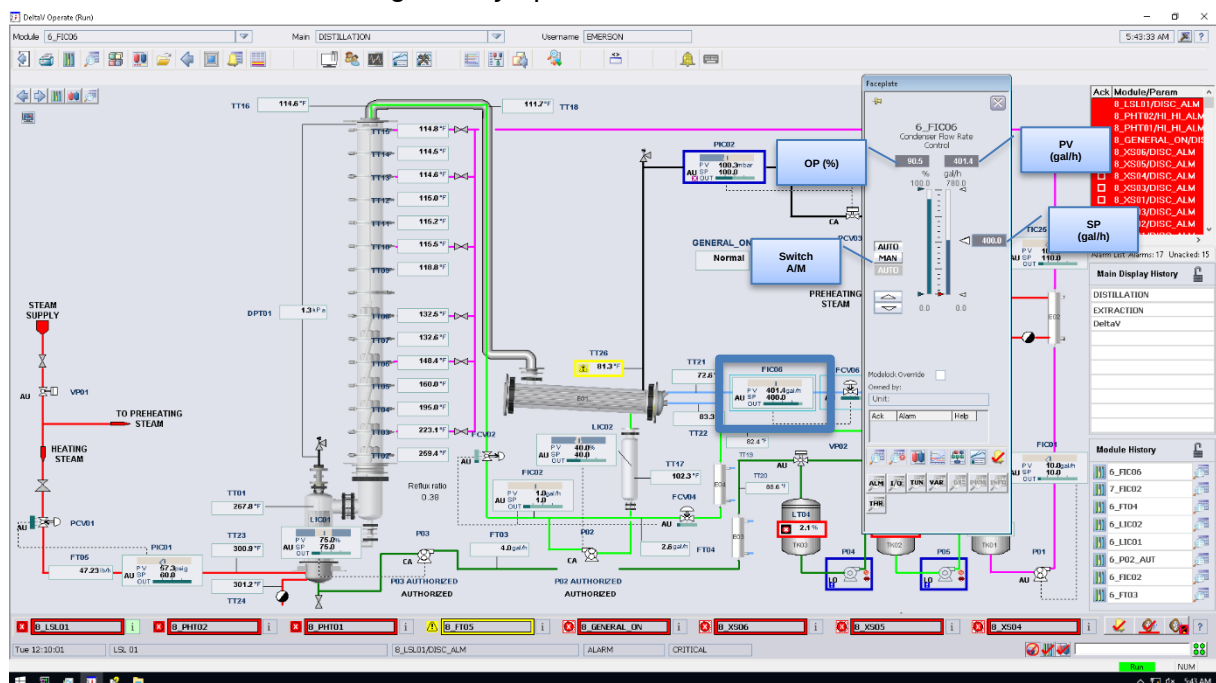
5.4.1. Water feeding through condenser E01 and coolers E03 to E07

Check that the hoses for steam condensate evacuation are oriented to the wasted water collection system of the laboratory.

Start the cooling water circulation, open V87, V89, V91 and V94, to feed the condenser and the whole coolers.

Click on FIC06 faceplate, switch on the Auto mode, enter the setpoint value SP 1 510 L/h (400 Gal/h).

The output value OP% will increase until you reach the setpoint target, that is to say the flow rate control valve FCV06 will gradually open and then stabilise.



FIC06: Condenser flow rate control faceplate

Remind:

If the cold-water flow rate is too low in comparison with the quantity of vapours to condensate, the general vent temperature will increase. The thermostat will then detect a high temperature alarm (40°C / 104°F) and stop the unit.

In this case:

Increase the water flow rate through the condenser E01 up to 1 510 L/h (400 Gal/h) to accelerate the cooling of the general vent thermostat.

In case of high temperature alarm on the thermostat, always think about the causes:

- Cold water flow rate not adjusted correctly.
- The unit has been put under reduced pressure.
- Restart the distillation unit as indicated in the paragraph "How to start the unit"

5.4.2. Column vacuum/nitrogen pressure control PIC02/P06

Check that the hoses from both extraction points are connected to the laboratory's extraction system.

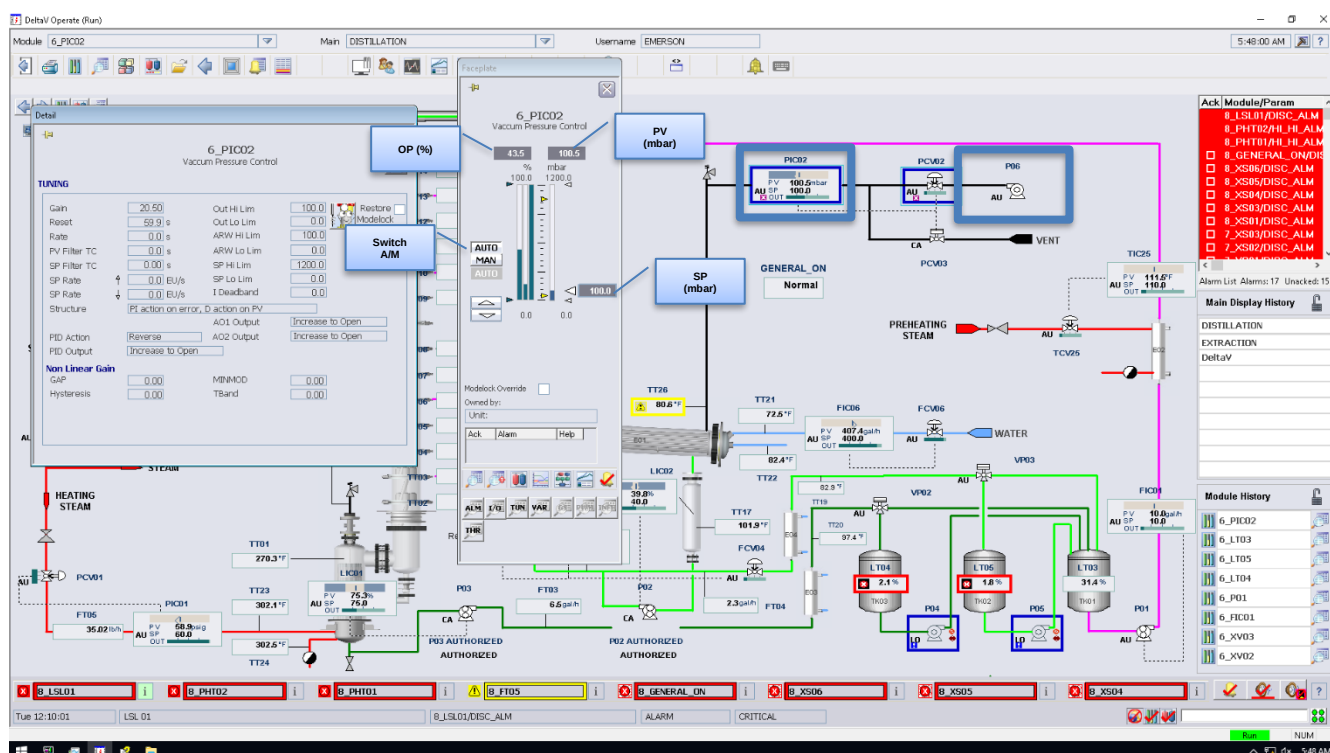
Click on PIC02 faceplate, switch on the Auto mode, enter the setpoint value SP 100 mbar (1.45 psi).

Pressure is controlled by two valves PCV02 and PCV03, but a single faceplate PIC02 for both. These valves are working in opposition, it means:

OUTPUT	0%	25%	50%	75%	100%
PCV02 (VACUUM)	Open	½ Open	Closed	Closed	Closed
PCV03 (NITROGEN)	Closed	Closed	Closed	½ Open	Open

Click on P06 faceplate to start the vacuum pump.

The PIC02 output value OP% will decrease until you reach the setpoint target, that is to say the pressure control valve PCV02 will gradually open and then stabilise.



PIC02: Vacuum pressure control faceplate

Remind:

If PIC02 process value doesn't decrease, check V77 is open, and V81 closed.

Check also that PCV02 has a sufficient AC pressure supply.

Check any other valve open through inadvertence (e.g., V109).

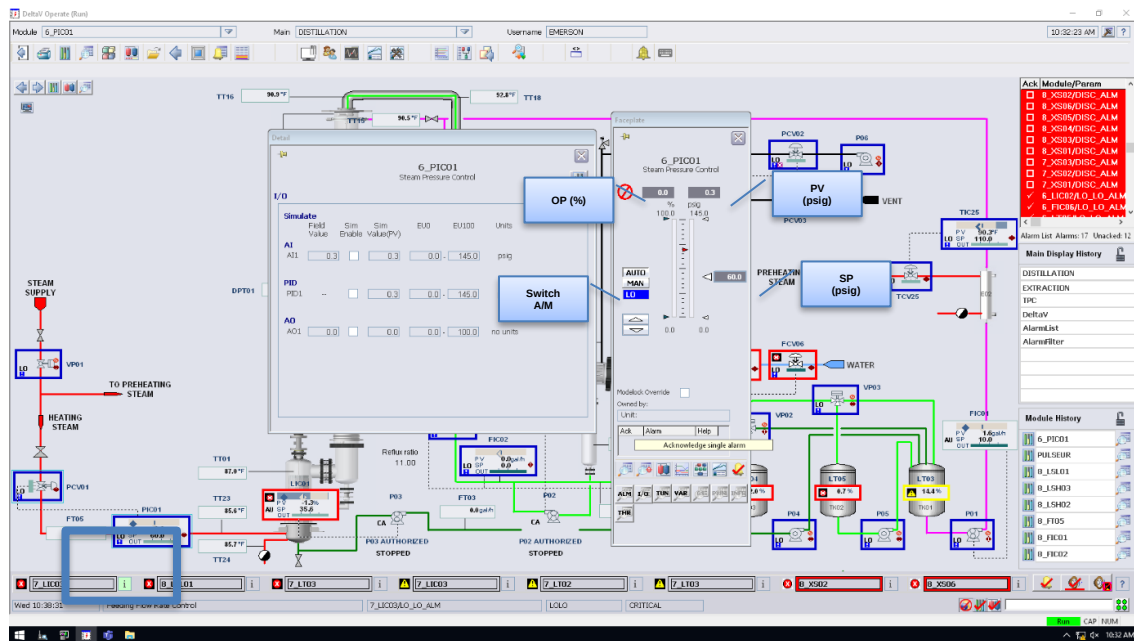
5.4.3. Boiler heating steam pressure control PIC01

Check that the steam condensate hoses are connected to the laboratory's draining point.
Check that the steam draining valve V110 is closed.

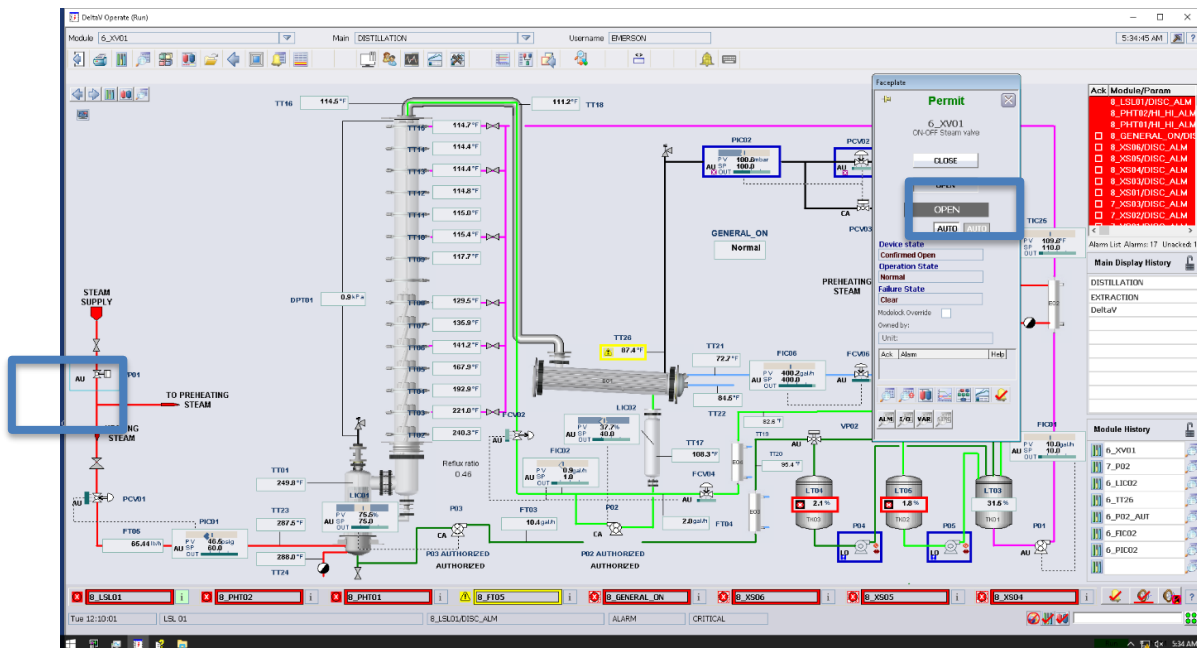
Click on PIC01 faceplate, switch on the manual mode, enter the output value OP 50%.
Pressure is controlled by PCV01, output increasing leads to steam pressure increasing.
Open V63 gate valve.

Click on VP01 faceplate and open it to start steam inlet.

Adjust the steam inlet pressure 4-5 barg (60-73 psig) with PRV01.



PIC01: Steam pressure control faceplate



VP01: Opening & Closing Steam valve faceplate

The steam mass flow rate is measured with a Vortex flow transmitter FIT05.
Wait until pressure stabilisation, then switch on auto mode and enter the setpoint value.

Remind:

We advise to use the manual control of the heating at the beginning of the experiment, until you observe vapours on top of the column. When the differential pressure value is near to the expected value 8-14 mbar (0.12-0.20 psi), then put the controller in automatic mode and indicate the pressure set point. The heating control is now automatic.

To stop heating:

Switch to PIC01 manual mode, set 0% output value to close the valve.
Then, close VP01.

5.4.4. Feed product flow rate FIC01/P01

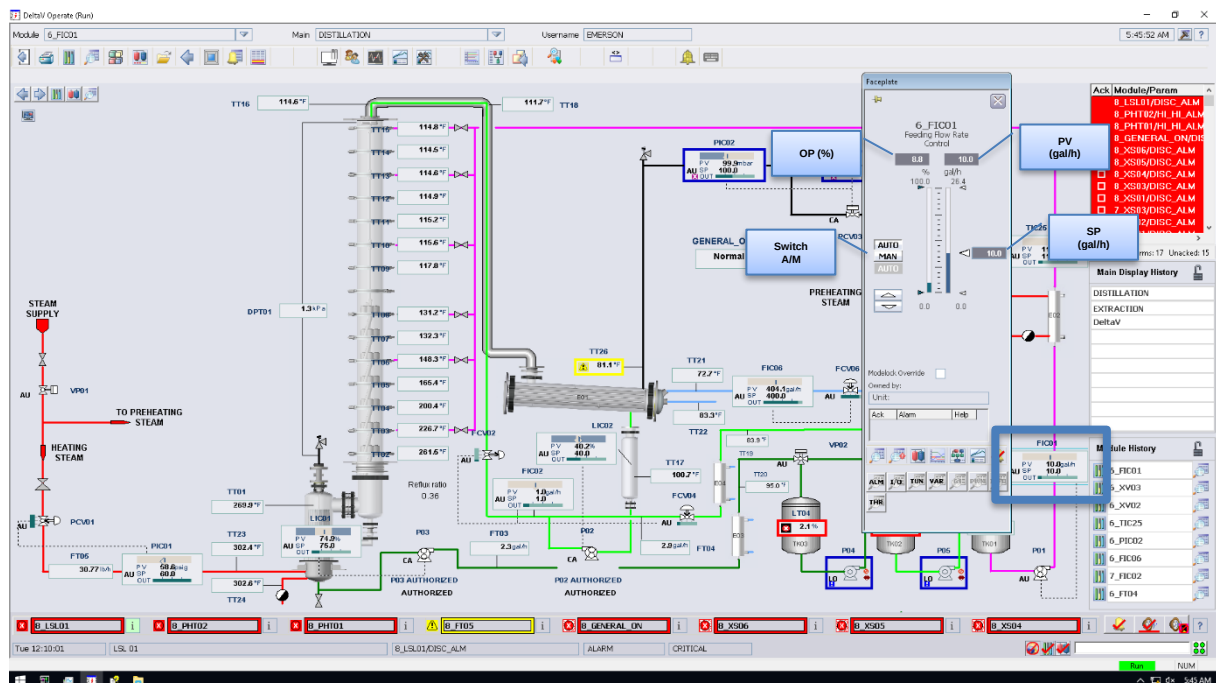
First check that any feed product valves have not been opened inadvertently (see section 5.2.4.).

Select a single column tray, and open its column feed product valve, either V12, or V13, or V14, or V15, or V16, or V17.

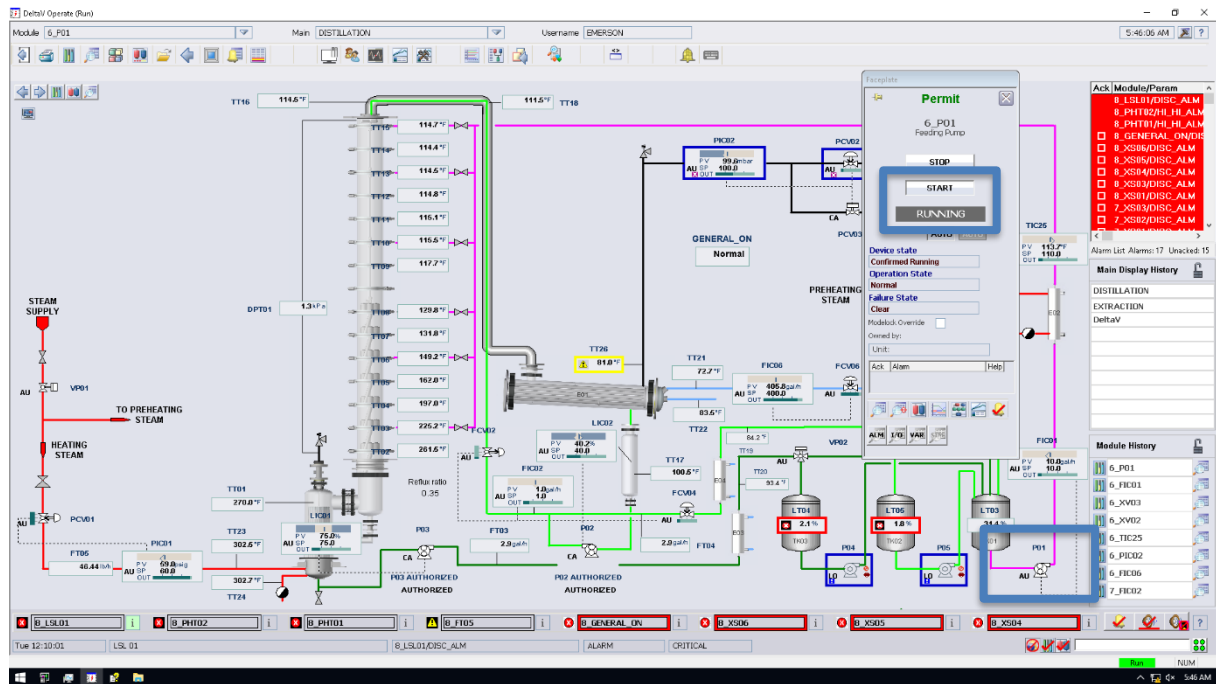
The feed flow rate is measured and controlled with a flow transmitter (FIC01) by means of a speed controller for the pump (P01).

Click on FIC01 faceplate switch on the Auto mode, enter the setpoint value SP 38 L/h (10.04 Gal/h).

Then click on P01 faceplate, start the pump.



FIC01: Feeding flow rate control faceplate



P01: Start & Stop Feeding pump faceplate

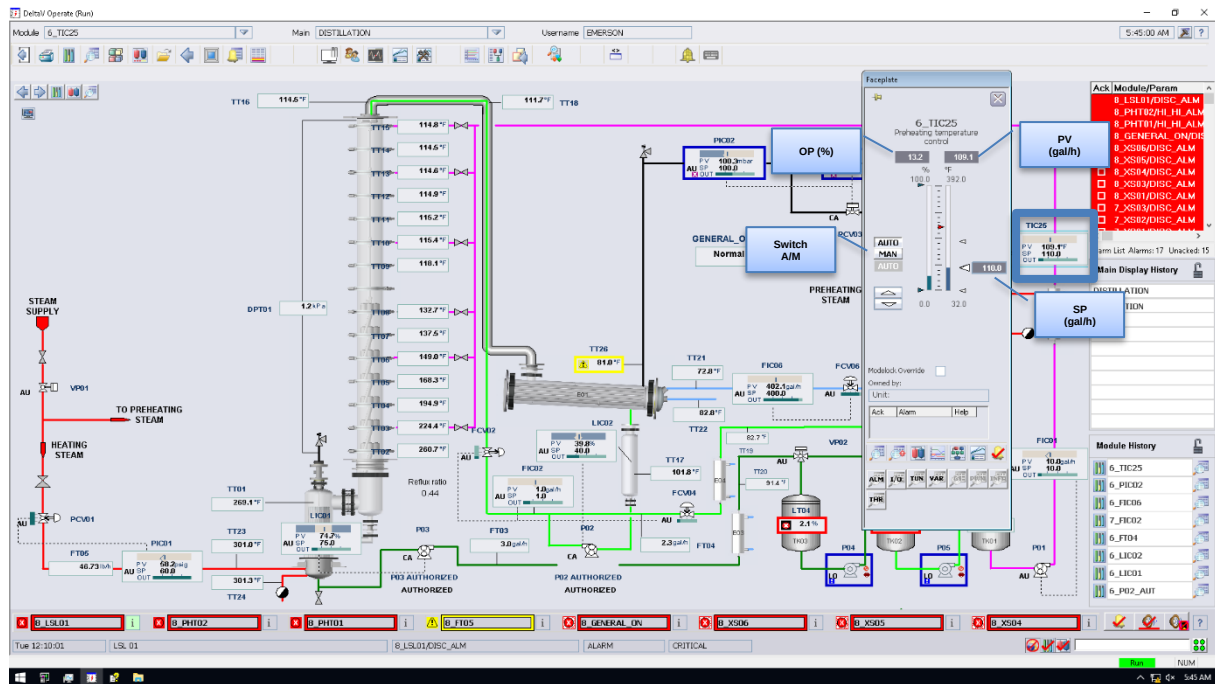
5.4.5. Feed product preheating temperature control TIC25

First check that any feed product valves have not been opened inadvertently (see section 5.2.4.). Check that the steam condensate hoses are connected to the laboratory's draining point.

Click on TIC25 faceplate, switch on the manual mode, enter the output value OP 20%. Temperature is controlled by TCV25, output increasing leads to preheating temperature increasing.

Open V63 gate valve. Click on VP01 faceplate and open it to start steam inlet. Adjust the preheating steam inlet pressure 0.69-1.38 bar (10-20 psig) with PRV02.

Wait until temperature stabilisation, then switch TIC25 on auto mode and enter the setpoint value corresponding to the column tray temperature. For instance, if you would like to introduce the feed product at 11th tray, as the temperature TT11 is 43.33°C (110°F), so enter a preheating temperature setpoint value TIC25 43.33°C (110°F).



TIC25: Preheating temperature control faceplate

5.4.6. Boiler level control LIC01/P03

The boiler level is measured with a differential pressure transmitter (LT01) and controlled by means of a speed controller for the pump (P03). The static pressure of the liquid height, that is to say the level through the boiler is expressed as a percentage.

Click on LIC01 faceplate, switch on the auto mode, enter the setpoint value SP 65-75-%.

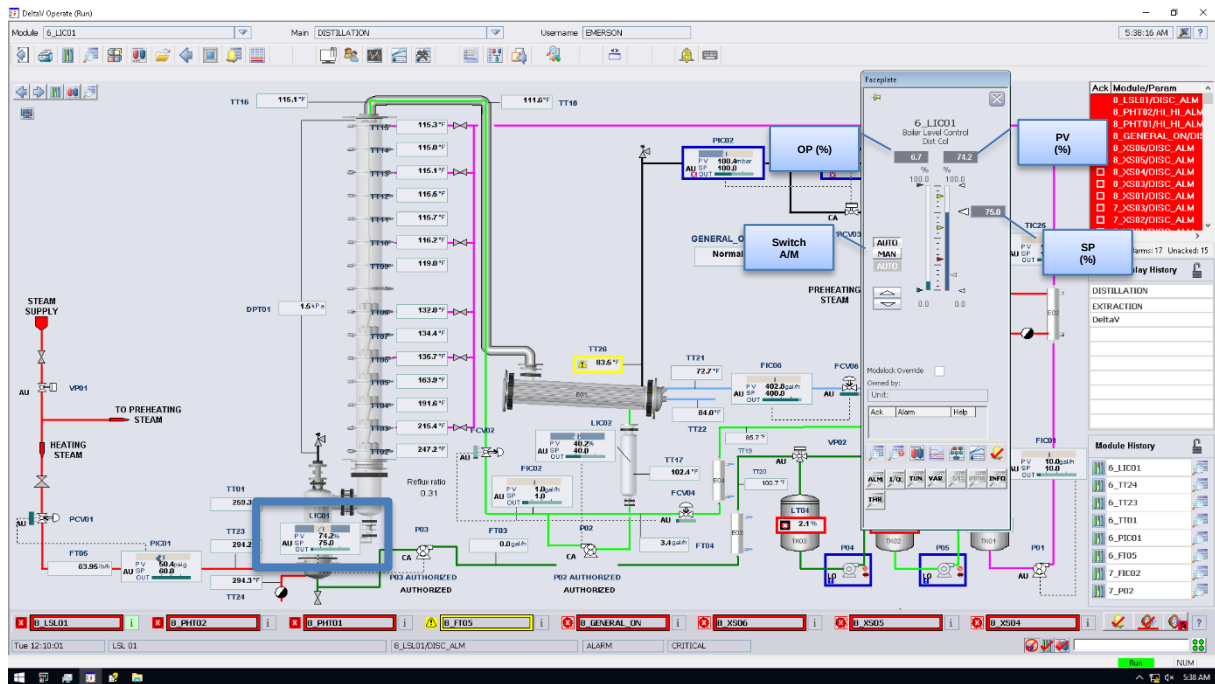
Wait until LIC01 reaches 76-79% (a slight variation), then click on P03_Authorized faceplate and authorize pump's functioning.

Click on P03 faceplate and authorize pump's functioning.

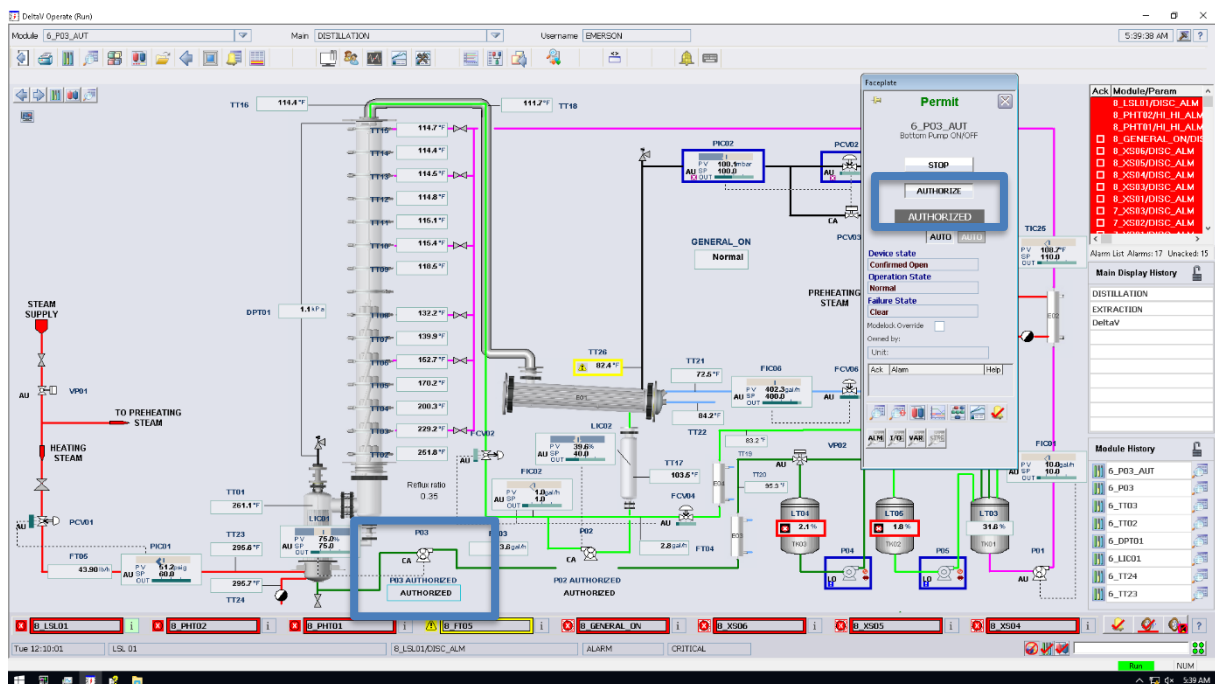
Click on VP02 faceplate, open/close VP02 to send the bottom product either in TK01 or TK03 respectively.

Remind:

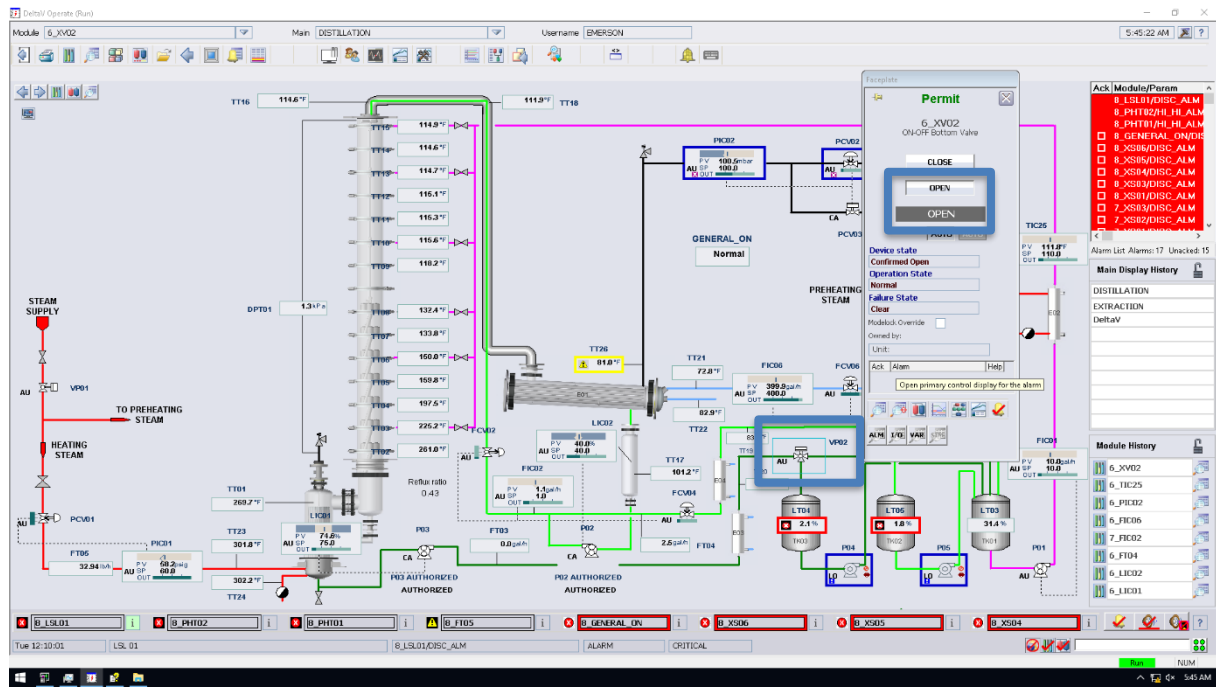
When P03's functioning is authorized, the pump remains on standby under a LIC01 threshold of 30%.



LIC01: Boiler level control faceplate



P03: Start & Stop Bottom pump faceplate



VP02: Opening & Closing Bottom product valve faceplate

5.4.7. Distillate level control LIC02/P02 and reflux flow control FIC02

The distillate level is measured with a differential pressure transmitter (LT02) and controlled by means of a speed controller for the pump (P02). The static pressure of the liquid height, that is to say the distillate receiver level is expressed as a percentage.

Reflux/Distillate flow is controlled by two valves FCV02 and FCV04, but a single control faceplate FIC02 for both. These two valves are working in opposition with a common Output, that is to say:

OUTPUT	0%	25%	50%	75%	100%
FCV02 (REFLUX)	Closed	¼ Open	½ Open	¾ Open	Open
FCV04 (DISTILLATE)	Open	¾ Open	½ Open	¼ Open	Closed

Click on LIC02 faceplate, switch on the auto mode, enter the setpoint value SP 40%.

Wait until LIC02 reaches 60-65%, then click on P02_Authorized faceplate and authorize pump's functioning.

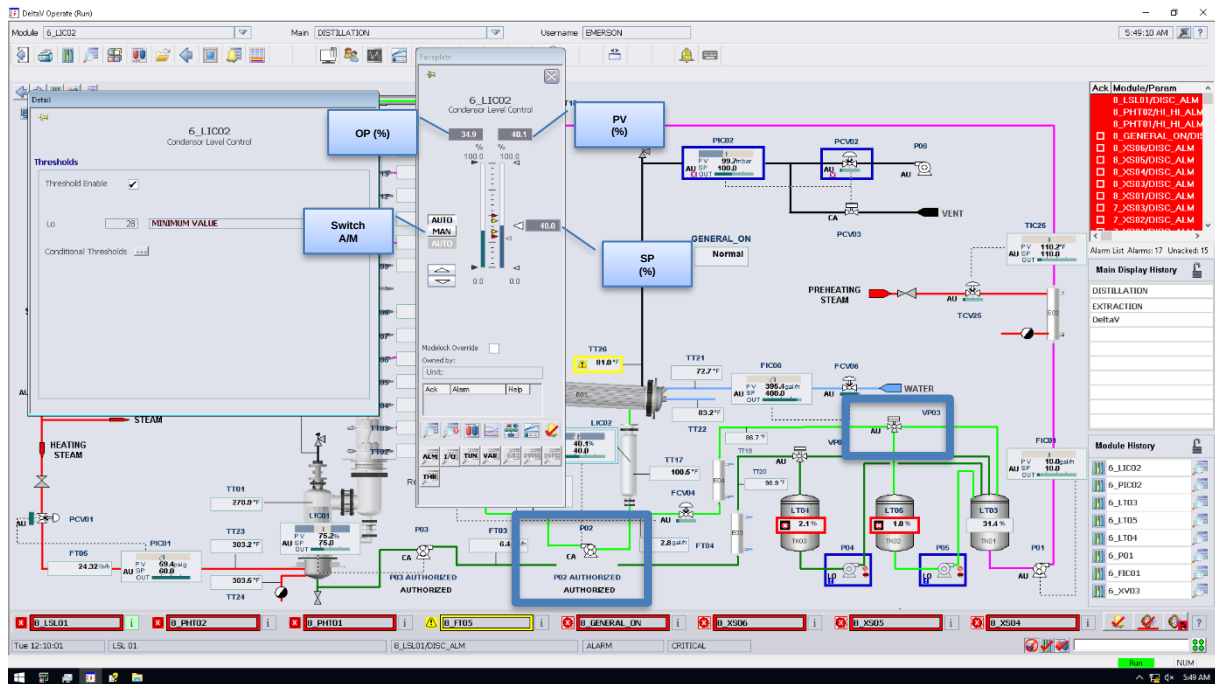
Click on FIC02 faceplate, switch on the auto mode, enter the reflux flow rate setpoint value SP 3.8 L/h (1 Gal/h).

Click on VP03 faceplate, open/close VP03 to send the distillate either in TK01 or TK02 respectively.

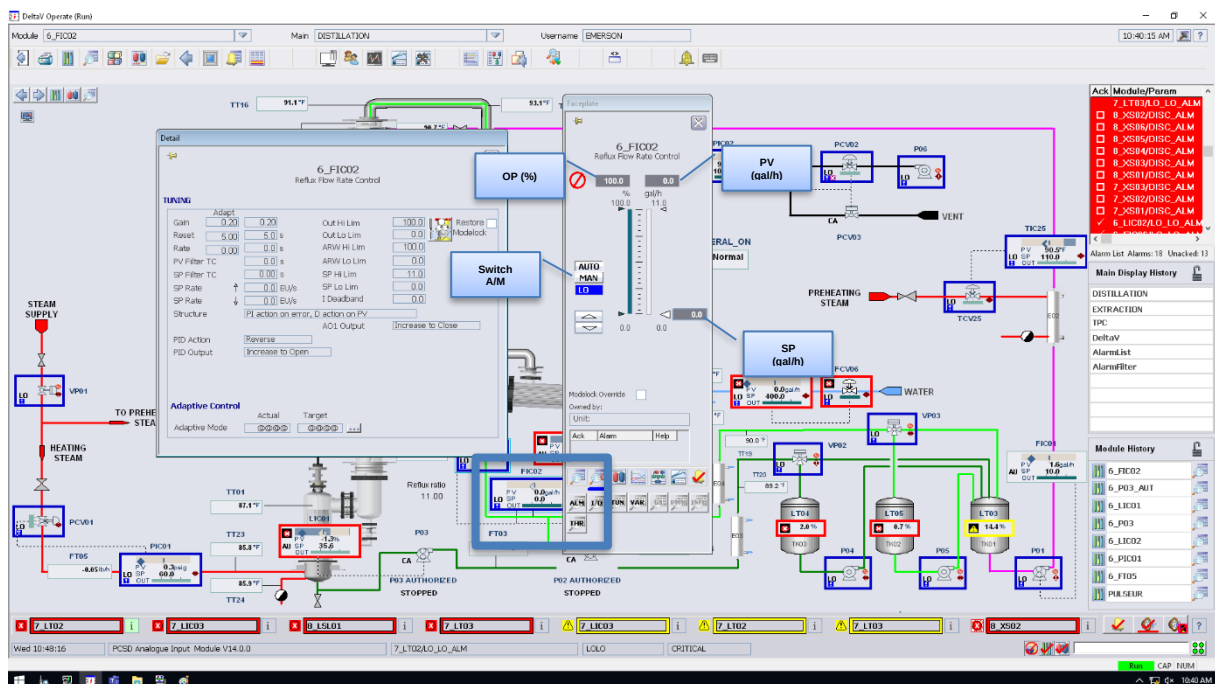
Remind:

When P02's functioning is authorized, the pump remains on standby under a LIC02 threshold of 28%.

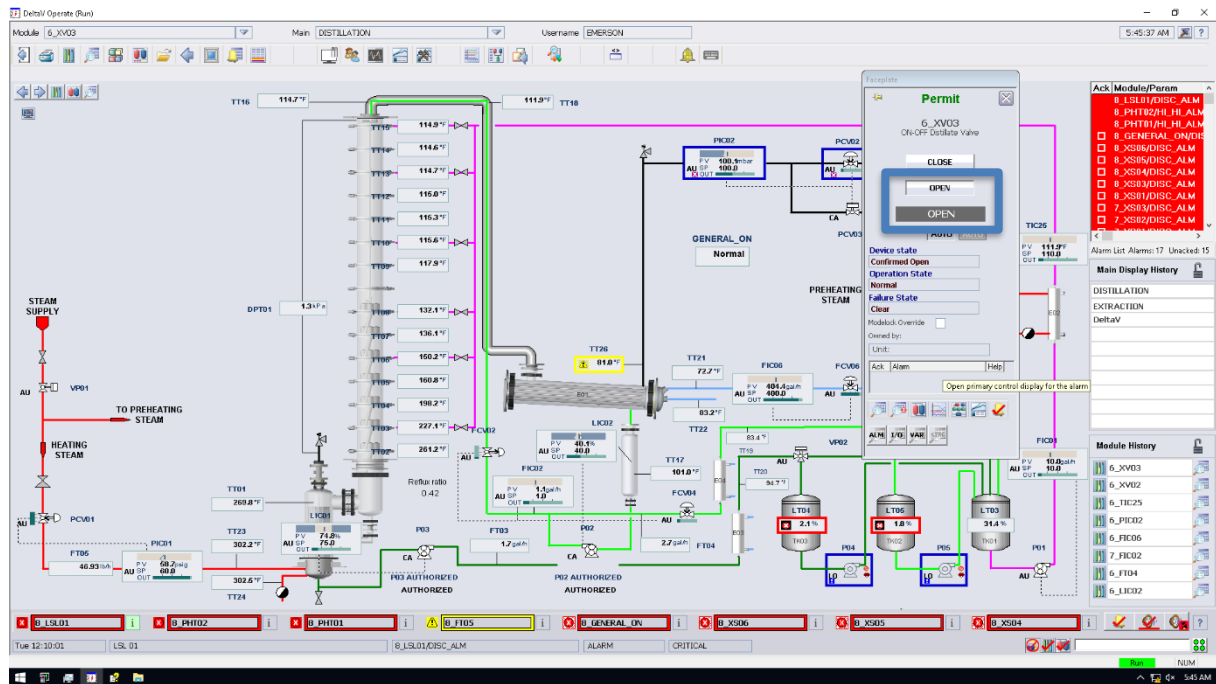
The opening of each control valve FCV02 and FCV04 can be seen on its own opening index.



LIC02: Distillate level control faceplate



FIC02: Reflux flow rate control faceplate



VP03: Opening & Closing Distillate valve faceplate

5.5. PROCEDURES FOR USE

5.5.1. Starting the vacuum continuous distillation

1. Fill up the feeding storage tank with 200 L of a 30% glycol mixture
2. Fill up the boiler with pure glycol until 65-75% (LIC01)
3. Feed the condenser with cold water circulation, at least 1 510 L/h (400 Gal/h).
4. Feed with cold water, distillate, bottom product and outlet steam coolers
5. Put the column under vacuum start pump P06 and adjust the process pressure PIC02 auto mode SP = 100 mbar (1.45 psi)
6. Wait for the pressure to stabilize at 100 mbar (1.45 psi)
7. Start heating the boiler until you note some vapours near the column reflux head: open PIC01 manual mode OP = 50% until you reach 3.45 barg (50 psig), then open VP01
8. Wait until the temperature profile along the column is stable
9. Select the feeding tray of the mixture into the column
10. Start the pump P01, feeding flow rate FIC01 auto mode SP = 38 L/h (10 Gal/h)
11. Adjust the feed preheating temperature TIC25 according to the column tray temperature selected, feeding preheating manual mode OP = 20%, then switch auto mode
12. Wait until a slight increase in LIC01 boiler level
13. Start the pump P3, LIC01 auto mode, SP = 65-75% to control the boiler level
14. When the distillate level LIC02 reaches 60%, set LIC02 auto mode SP = 40%, then start the pump P2
15. Set the reflux flow rate, FIC02 in auto mode SP = 3.8 L/h (1 Gal/h)
16. Wait for all temperatures to stabilize
17. Write down all process values and set points
18. Take samples of the interesting phases, for instance feeding mixture, distillate and residue
19. Follow the experiment, changing the distillation parameters:
 - Reflux ratio
 - Feeding tray into the distillation column.
 - Preheating temperature
 - Condenser cold water flow rate

5.5.2. Taking samples

5.5.2.1. Feed product

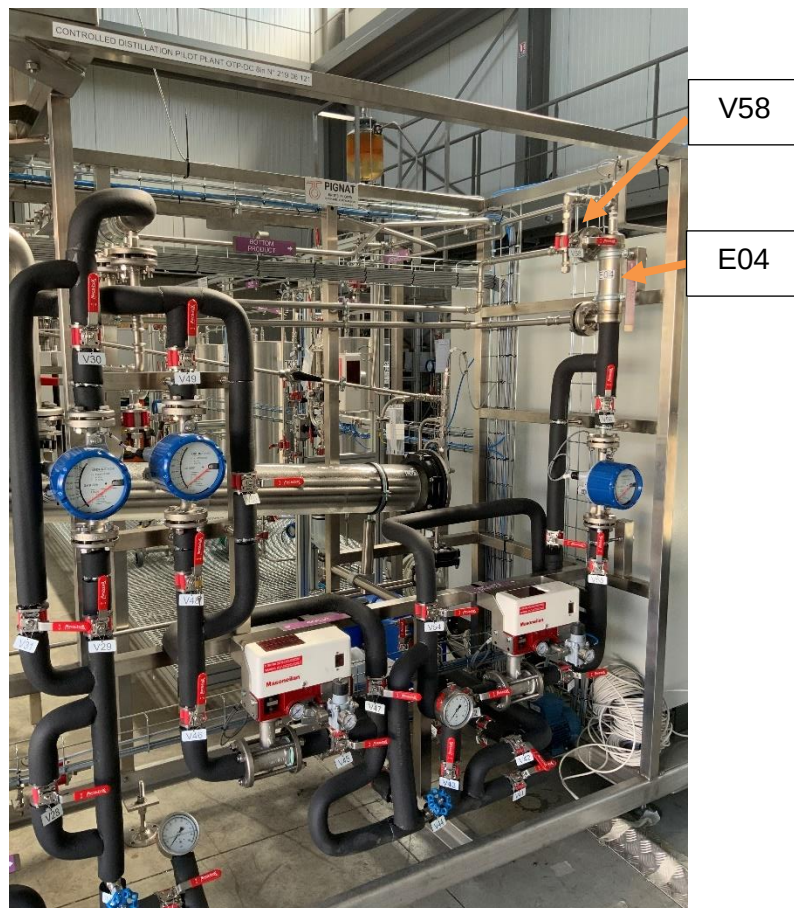
Feeding product can be mixed in the tank, starting P01 and opening V108 by-pass valve back to TK01. Open V1 to take a sample.



TK01: Feeding tank

5.5.2.2. Distillate product

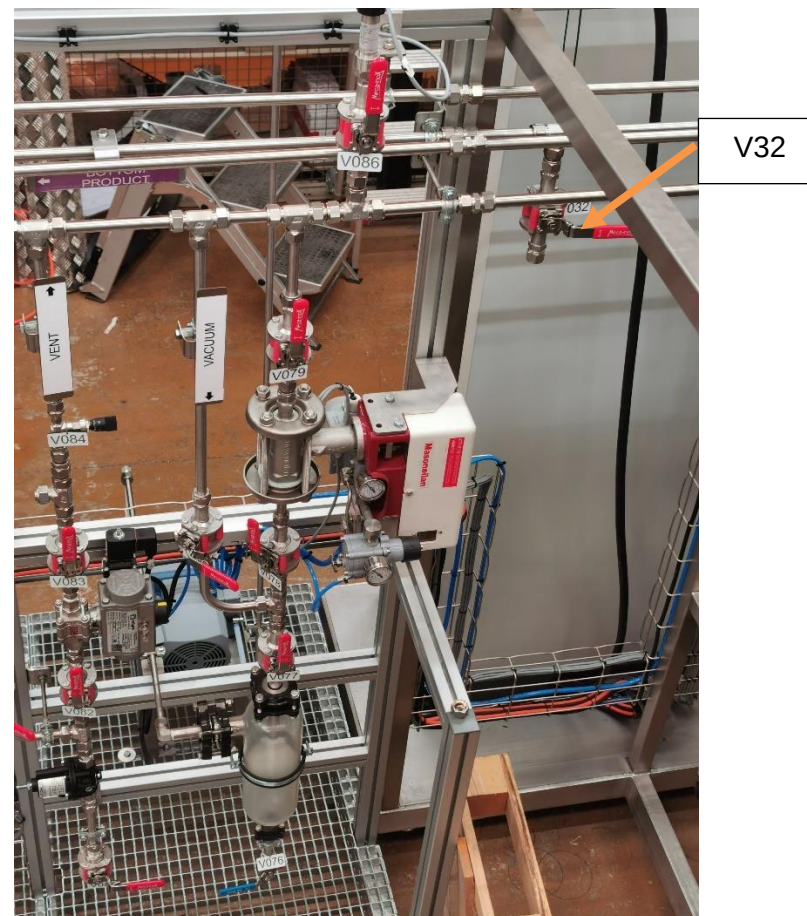
Distillate product is cooled down through E04
Open V58 to take a sample.



V58: Distillate product sampling valve location

5.5.2.3. Bottom Product

Bottom product is cooled down through E03
Open V32 to take a sample.



V32: Bottom product sampling valve location

5.5.3. Experimental characterization of the process

Write down the following values:

- Temperatures from TT01 to TI18, differential pressure drops DPT01
- Flow rates: cold water in condensers FT06, column feeding flow rate FT01, distillate and residue flow rates FT02/FT03/FT04...
- Constant parameters: feeding tray into the column...
- Take samples of the following products: feeding mixture (Open V1), bottom product (Open V32), and distillate (Open V58)

For each phase, measure at least the density, and use the data table below, for determining the glycol concentration. You can, if possible, determine the composition of each sample using adapted analytical method, depending on the process.

		<i>Temperature of MEG/Water solution</i>							
<i>Concentration of glycol (MEG) in percentage by weight of solution</i>		<i>10°C</i>	<i>20°C</i>	<i>30°C</i>	<i>40°C</i>	<i>50°C</i>	<i>60°C</i>	<i>70°C</i>	<i>80°C</i>
		<i>50°F</i>	<i>68°F</i>	<i>86°F</i>	<i>104°F</i>	<i>122°F</i>	<i>140°F</i>	<i>158°F</i>	<i>176°F</i>
	<i>0 %</i>		1000	995	992	988	983	978	972
	<i>10%</i>	1013	1010	1008	1005	1000	996	990	985
	<i>20%</i>	1028	1025	1020	1017	1012	1008	1000	995
	<i>30%</i>	1042	1039	1034	1030	1023	1018	1011	1005
	<i>40%</i>	1057	1051	1047	1041	1035	1030	1023	1018
	<i>50%</i>	1070	1066	1060	1054	1048	1041	1035	1028
	<i>60%</i>	1083	1077	1070	1065	1058	1051	1045	1038
	<i>70%</i>	1095	1088	1082	1075	1069	1062	1055	1047
	<i>80%</i>	1105	1099	1091	1085	1078	1070	1064	1057
	<i>90%</i>	1113	1107	1100	1092	1085	1078	1071	1064
	<i>100%</i>	1120	1114	1107	1100	1092	1085	1078	1070

MEG/Water solution density (kg/m^3) as a function of solution temperature and concentration

5.5.4. Data analyse

Mass balance

Calculate the global mass balance by means of the flow transmitters (FT01, FT02, FT03 and FT04)

Calculate the partial mass balance concerning the volatile compound.

Thermal balances:

Condenser

Calculate the thermal power exchanged between the cold fluid (water circulation into the tubes) and the vapours while condensing. Compare these two values.

Boiler

Calculate the energy balance on steam: with the following parameters:

- Steam flow rate FT05 in kg/h (lb/h)
- Inlet steam temperature in the re-boiler TT23
- Outlet steam temperature in the re-boiler TT24
- By steam condensate (filling rate of a graduated capacity)

Compare this value to the value obtained at the condenser, concerning the cold fluid thermal balance

5.6. STOPPING THE UNIT

5.6.1. Stopping during an experiment

When the unit is in use, it can be easily stopped:

5.6.1.1. Emergency shut-down

In case of an emergency, push the emergency circuit-breaker (mushroom-type).

In this case, pumps will be stopped as well as heating and stirring.

The voltage remains in the electrical cabinet but the control circuit is no longer powered.

To restart the unit, the user has to solve the problem that led to the emergency stop.

After making sure that the installation can be safely restarted, use the key to unlock the emergency stop.

5.6.1.2. Normal shut-down

During operation, the user can easily:

- Stop the heating, close VP01, PCV01 and TCV25.
- Stop the pumps
 - P1 feeding pump
 - P3 bottom product pump
 - P2 distillate pump lastly, to avoid any accumulation in R01 distillate glass receiver
- Break the process vacuum, open PCV02 in manual mode OP = 100%
- Stop the circulation of cold water, close FCV06 in manual mode OP = 0%

Remind:

P02 and P03 run in authorization mode, that is to say under a low threshold level, they will switch from on to standby.

So, it means that you could let these two pumps running until you reach that low threshold.

But at the end of an experiment, if you are sure to operate the same one on next time, you should let the boiler filled with pure glycol, and drain only the distillate glass receiver R01.

Stop the cold water lastly, according to the safety interlock file, when FCV06 is closed, it means no cold-water flow, all power devices switch on fallback position.

5.6.2. Shut-down

At the end of the experiment, the unit will always be cooled to room temperature before stopping it:

- Stop heating.
- Open FCV06 in manual mode OP = 100%
- Read the different temperatures.
- When temperatures are below 60°C (140°F), stop the circulation of cold water by closing FCV06.
- Stop the pump(s) by pressing the STOP button(s).
- Drain the unit.
- Clean the unit.
- Switch off the unit: switch the main switch to "0".
- Close the main sectional valve on the tap water network connected to the unit, and the steam inlet gate valve V63.

The unit is thus left out of energy.

WARNING

When an installation is shut down, it must be left out of energy, i.e. all supplies are closed (water, electricity, compressed air, steam...).

5.7. EMPTYING THE UNIT

5.7.1. Emptying during or at the end of an experiment

If you are distillate a binary glycol/water mixture then you can recycle the bottom product and distillate to prepare a new feeding mixture.

During the distillation: empty both receiver tanks **at the same time** in the feeding tank by means of the pump P04 and P05.

At the end of the experiment: empty both receiver tanks into the feeding tank and check that the mixture composition is correct.

You can also empty completely the unit and recycle all products into the feeding tank. Of course, we advise to take a sample of the mixture, measure density and adjust the composition, if necessary, with fresh products.

We advise to empty completely the unit between two experiments to avoid the unit to become dirty (possible deposit of impurities, especially if you are using technical quality products).

Nevertheless, if you intend to distillate the same mixture during the same week, it is not necessary to clean the unit.

If you intend to distillate a different mixture for the next experiment, then you have to clear the unit and clean it with adequate solvent. For example, carry out the distillation of the solvent alone, in order to clean all pipes and parts of the distillation unit. Use demineralised water in the case of ethanol and water mixture distillation.

Reminder:

Emptying, clearing and cleaning the unit are part of the workshop.

5.7.2. Checking before draining

Power off the unit using the main switch or the emergency stop button.

Check that the boiler temperatures TT01 is lower than 30°C (86°F).

5.7.3. Draining the plates of the column

Take care: you only have to empty the plates if you intend to distillate a mixture different than the one you processed during the workshop.

All plates will empty naturally when the pressure in the column drops and when the unit is cooling slowly, always with the boiler tank closed (do not open the charging valve). In fact, all vapours located on the liquid surface create a pressure drop at the bottom of the column. This pressure drop makes flowing down the liquid from upper trays to lower trays then to the boiler.

How to proceed?

Stop the unit.

Let the unit cool down to room temperature lower than 30°C (86°F) (depending of the season).

5.7.4. Boiler draining

Temperature TT01 must be lower than 30°C (86°F).

Place a draining flexible hose from V22 into a storage tank of feeding mixture.

Open this valve to empty completely the Boiler.

Do not forget to close the valve when draining is finished.

5.7.5. Receivers and Crash tank draining

Transfer the distillate and bottom product from TK02 and TK03 tanks into the feeding storage tank TK01 using P04 and P05 pumps.

Then place all draining flexible hoses from V33 and V59 into the storage tanks for bottom product and distillate.

Open these valves to empty completely the receivers and the crash tank.

Do not forget to close the valves when draining is finished.

5.7.6. What to do with the products used during the experiments

The unit operates with water and glycol that will not be discharged to the sewer.

During cleaning operations (in case of limescale deposits or bacterial development), the discharge of these cleaning waters must be carried out according to the rules in force in the institution.

5.8. CLEANING THE UNIT

WARNING

Do not use strong detergents or oxidizing agents (e.g. bleach).

5.8.1. Cleaning the outside of the unit

The frame and the various parts will be cleaned with a mild detergent.
The assembly will be rinsed with water then wiped after use with a soft cloth.

5.8.2. Cleaning the inside of the unit

Since the unit operates with water and glycol, there is no specific cleaning at the end of handling. It is advisable to drain it daily or weekly to avoid limescale deposits and bacterial development.

In the case of limescale deposits, clean the inside of the unit with a dilute acid solution. These dilute solutions (5-10% of acetic acid or nitric acid) are manually introduced into TK01, TK02, TK03 and TK04 using a funnel.
The operator must wear his PPE and work in compliance with the principles of good laboratory practice (GLP).

After acid cleaning, the unit will be thoroughly rinsed until full neutrality even in "dead volumes" (pump heads, pressure tapings...).

Anti-algae or anti-bacterial products can also be used to clean the unit or to store water in it. These products must be compatible with the materials of the unit.
After use, the unit will be thoroughly rinsed with water.

At the end of cleaning, the unit is rinsed then emptied.

5.9. PRACTICAL NOTEBOOK

There must be a practical notebook with each unit. This is set up by the person in charge of the hall/lab. It is to be filled in at the end of each experiment carried out on the unit. It must contain the following information:

- Date
- Name of the person responsible for the practical session
- Name of the operator
- Type of reagent and quantity used
- Type of synthesis done as well as the operating conditions
- Verification of the unit at the end of the experiment:
- Emptied, rinsed and clean inside and out
- Comments about any possible failures on the unit.

5.10. STORING THE UNIT

During long periods when the unit is not used, it is important that the unit is stored correctly, i.e.:

- Flushed through, emptied, cleaned and dried
- Disconnected from the mains water, electricity, and compressed air supplies, etc.
- Put the unit in a closed and clean place, where it is sheltered from the weather, and away from direct sunlight, dust, solid particles and corrosive products, etc.
- Store the unit in the temperature conditions specified on the unit label and/ in this manual.

DISTILLATION PROCESS OPERATION

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6. *UPKEEP AND MAINTENANCE*

6.1. GUARANTEE

The standard guarantee for PIGNAT equipment is as follows:

- 1-year guarantee on all parts and workmanship against any hidden defects.
- Start of the guarantee is from the date the equipment arrives on the client's site.

Refer to the quote and order if any special conditions apply.

The guarantee only covers the replacement or repair of the parts recognized as being defective. The modified parts or those replaced do not have an extended guarantee above that given by the standard guarantee period of the main unit.

All the defective parts must be replaced by those certified as original and identical to those specified by the manufacturer. The list of parts making up the unit is given in the technical file delivered with this unit.

Only the manufacturer or persons with express rights are qualified to ensure the repair or replacement of defective parts.

The limits of the 1-year guarantee on all parts and workmanship against any hidden defects are given hereafter.

The equipment corresponds to the operating conditions and uses defined and brought to our attention when the unit was ordered.

Any damage due to conditions of use which are non-conform or do not respect the specified instructions in this manual are excluded from the guarantee.

Other aspects excluded from the guarantee are:

- the replacement of PIGNAT parts by those from another company or origin
- cleaning, maintenance, verification, and inspection
- assembly of external parts or modification of the construction of the unit which have not been validated by PIGNAT
- parts which have become worn.

6.2. LIST OF SPARE PARTS

The nomenclature goes with the PID where the parts are indicated.

The list of spare parts and the PID can be seen with the references in the technical file accompanying this unit.

6.3. TECHNICAL FILE FROM THE MANUFACTURER

The technical document for each element on the unit is given in the technical file supplied when the unit is delivered to the client or on initial operation.

It is the client's responsibility to read these documents in order to know the specifications of each manufacturer for the maintenance of each element.

It is also the client's responsibility to give each person working on this unit the necessary information in order to complete the tasks on the unit i.e. operation, cleaning, maintenance, inspection, etc.

6.4. MAINTENANCE LOGBOOK

From initial operation and for the whole life-time of the unit, the client must provide a maintenance logbook for the unit in which are recorded all the operations and interventions done on the unit with respect to cleaning, maintenance, and inspection etc.

The maintenance logbook must be provided for each unit by the person in charge of it.

It is to be filled in after each intervention, even a visual inspection, by the person responsible for this intervention.

The maintenance logbook must contain the following information:

- Date
- Name of the person doing the operation – cleaning, inspection, maintenance, etc.
- The type of intervention or maintenance: general or preventive, etc.
- Description of the cleaning or maintenance operations realized.

6.5. MAINTENANCE OPERATIONS

It is highly recommended that the maintenance on the unit is done by a trained technician designated by the PIGNAT company.

If the person realizing the intervention has not been designated by PIGNAT, then in order to limit any risk, the client must ensure that this person is qualified to do the task. The client must supply all the necessary documents, tools and equipment in order to accomplish the task(s).

Before any intervention on the unit, it is essential to ensure that the unit is at atmospheric pressure (no vacuum, nor under pressure, etc.), that it does not contain any liquid, is clean and correctly ventilated, and that all the elements are at ambient/room temperature and finally, no residual risk exists.

As well as the procedures described below, the pilot unit must always be kept in a good state and checked as often as necessary.

The observations and periodic inspections made must conform to the regulations applicable in the country where the unit is installed. If there are no regulations, then we recommend the following:

- Verification that the connections are airtight and watertight for each operation of the unit
- A weekly visual inspection
- A maintenance visit every year.

Inspection of the safety elements, inspections inside the unit (notably for corrosion) and making sure that the equipment meets the specifications must all respect the regulations applicable in the country where the unit is being used, and, if there are no such regulations, be defined by the client according to the operations done on the pilot unit.

6.5.1. Pressure reducer

Pressure reducer are equipped with a filter and a condensate recovery tank.

In case of presence of condensates:

- Open the cap at the bottom of the tank.
- Let the water flow.
- Close the cap.

In case of clogging or deterioration of the filter, replace it.

6.5.2. Sealing

In the event of a leak at a fitting, immediately replace the seal or sealing tape with an identical element.

If this is not possible, stop the unit and place a warning sign on it.

6.5.3. Electrical control

This control is to be carried out by authorized electrical personnel on non-energy logged equipment.

Weekly, check that:

- The cable glands on the cabinet and the instruments are correctly tightened.
- No electrical cable is pinched or damaged.

Annually, check the tightness of the terminals in the control cabinet.

In the event of damage to a cable or an electrical element, replace it immediately with an identical element.

If this is not possible, stop the unit and place a warning sign on it.

DISTILLATION PROCESS OPERATION

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7. APPENDICES

7.1. PHYSICAL CHARACTERISTICS OF WATER-GLYCOL MIXTURE

MEG 30%

solution aqueuse de monoéthylène glycol à 30%
30% monoethylene glycol aqueous solution

température de fusion : -14.5°C
freezing temperature

température θ temperature	masse volum. ρ density	chaleur spécifique Cp specific heat		viscosité dynamique μ dynamic viscosity		conductivité thermique λ thermal conductivity		pression de vapeur Ps vapor pressure	
°C	kg/m3	J/kg.K	kcal/kg.°C	Pa.s	kg/m.h	W/m.K	kcal/h.m.°C	Pa (abs)	bar(rel.) barg
-10	1057	3611	0,863	0,003465	25,908	0,464	0,399	254	
-5	1056	3620	0,865	0,002931	20,729	0,467	0,402	374	
0	1054	3629	0,867	0,002501	16,852	0,470	0,405	543	
5	1052	3639	0,869	0,002151	13,902	0,473	0,407	775	
10	1050	3650	0,872	0,001864	11,625	0,476	0,409	1092	
15	1048	3661	0,875	0,001626	9,843	0,478	0,411	1517	
20	1046	3672	0,877	0,001428	8,431	0,480	0,413	2081	
25	1044	3684	0,880	0,001262	7,299	0,482	0,415	2821	-0,99
30	1042	3696	0,883	0,001122	6,381	0,484	0,416	3780	-0,98
35	1040	3708	0,886	0,001003	5,630	0,486	0,418	5011	-0,96
40	1037	3721	0,889	0,000902	5,010	0,487	0,419	6574	-0,95
45	1035	3733	0,892	0,000814	4,493	0,489	0,420	8541	-0,93
50	1032	3746	0,895	0,000739	4,059	0,490	0,421	10995	-0,90
55	1030	3759	0,898	0,000674	3,692	0,491	0,422	14031	-0,87
60	1027	3772	0,901	0,000617	3,380	0,492	0,423	17755	-0,84
65	1025	3785	0,904	0,000567	3,112	0,493	0,424	22290	-0,79
70	1022	3798	0,907	0,000523	2,881	0,494	0,425	27772	-0,74
75	1020	3811	0,911	0,000485	2,681	0,495	0,426	34353	-0,67
80	1017	3824	0,914	0,000451	2,507	0,496	0,426	42203	-0,59
85	1015	3837	0,917	0,000420	2,355	0,496	0,427	51508	-0,50
90	1012	3850	0,920	0,000393	2,221	0,497	0,427	62472	-0,39
95	1009	3863	0,923	0,000369	2,103	0,497	0,428	75318	-0,26
100	1007	3876	0,926	0,000347	1,999	0,498	0,428	90290	-0,11
105	1004	3888	0,929	0,000328	1,906	0,498	0,429	107649	0,06
110	1002	3901	0,932	0,000310	1,824	0,499	0,429	127677	0,26
115	999	3914	0,935	0,000294	1,750	0,499	0,429	150678	0,49
120	997	3926	0,938	0,000279	1,684	0,499	0,430	176975	0,76
125	994	3939	0,941	0,000266	1,625	0,500	0,430	206913	1,06
130	992	3951	0,944	0,000254	1,572	0,500	0,430	240858	1,40
135	989	3963	0,947	0,000243	1,524	0,500	0,430	279198	1,78
140	987	3975	0,950	0,000233	1,481	0,500	0,430	322341	2,21
145	984	3987	0,953	0,000224	1,442	0,500	0,430	370717	2,69
150	982	3999	0,955	0,000215	1,407	0,501	0,431	424776	3,23
155	979	4011	0,958	0,000208	1,375	0,501	0,431	484993	3,84
160	977	4023	0,961	0,000200	1,347	0,501	0,431	551859	4,51
165	975	4034	0,964	0,000194	1,321	0,501	0,431	625889	5,25
170	972	4046	0,967	0,000188	1,297	0,501	0,431	707618	6,06
175	970	4057	0,969	0,000182	1,276	0,501	0,431	797601	6,96



MEG 40%

solution aqueuse de monoéthylène glycol à 40%
40% monoethylene glycol aqueous solution

température de fusion : **-23.3°C**
freezing temperature

température θ temperature	masse volum. ρ density	chaleur spécifique Cp specific heat		viscosité dynamique μ dynamic viscosity		conductivité thermique λ thermal conductivity		pression de vapeur Ps vapor pressure	
°C	kg/m3	J/kg.K	kcal/kg.°C	Pa.s	kg/m.h	W/m.K	kcal/h.m.°C	Pa (abs)	bar(rel.) barg
-20	1079	3380	0,808	0,006124	61,722	0,435	0,375	104	
-15	1077	3393	0,811	0,005047	47,090	0,437	0,376	160	
-10	1076	3406	0,814	0,004201	36,628	0,438	0,377	239	
-5	1074	3420	0,817	0,003530	29,001	0,440	0,378	353	
0	1072	3435	0,821	0,002992	23,340	0,441	0,379	511	
5	1070	3450	0,824	0,002557	19,069	0,441	0,380	731	
10	1067	3465	0,828	0,002202	15,797	0,442	0,380	1029	
15	1065	3481	0,832	0,001911	13,255	0,443	0,381	1430	
20	1063	3497	0,835	0,001669	11,255	0,443	0,381	1961	
25	1060	3513	0,839	0,001468	9,661	0,443	0,381	2658	-0,99
30	1057	3529	0,843	0,001299	8,378	0,443	0,381	3562	-0,98
35	1055	3545	0,847	0,001156	7,334	0,444	0,381	4722	-0,97
40	1052	3561	0,851	0,001034	6,476	0,444	0,381	6196	-0,95
45	1049	3578	0,855	0,000931	5,765	0,443	0,381	8050	-0,93
50	1047	3594	0,859	0,000841	5,171	0,443	0,381	10364	-0,91
55	1044	3611	0,863	0,000764	4,671	0,443	0,381	13226	-0,88
60	1041	3627	0,867	0,000697	4,247	0,443	0,381	16738	-0,85
65	1038	3643	0,870	0,000639	3,885	0,442	0,381	21014	-0,80
70	1035	3659	0,874	0,000588	3,574	0,442	0,380	26184	-0,75
75	1033	3675	0,878	0,000543	3,305	0,442	0,380	32393	-0,69
80	1030	3691	0,882	0,000503	3,072	0,441	0,380	39798	-0,62
85	1027	3707	0,886	0,000468	2,869	0,441	0,379	48578	-0,53
90	1024	3723	0,890	0,000437	2,691	0,440	0,379	58925	-0,42
95	1021	3739	0,893	0,000409	2,534	0,440	0,378	71050	-0,30
100	1018	3754	0,897	0,000384	2,395	0,439	0,378	85183	-0,16
105	1016	3770	0,901	0,000362	2,272	0,439	0,377	101573	0,00
110	1013	3785	0,904	0,000341	2,163	0,438	0,377	120487	0,19
115	1010	3800	0,908	0,000323	2,065	0,437	0,376	142211	0,41
120	1007	3815	0,912	0,000307	1,978	0,437	0,376	167055	0,66
125	1005	3830	0,915	0,000292	1,899	0,436	0,375	195344	0,94
130	1002	3845	0,919	0,000278	1,829	0,435	0,374	227427	1,26
135	999	3859	0,922	0,000266	1,765	0,435	0,374	263670	1,62
140	997	3874	0,925	0,000255	1,708	0,434	0,373	304464	2,03
145	994	3888	0,929	0,000244	1,656	0,433	0,373	350216	2,49
150	991	3902	0,932	0,000235	1,609	0,433	0,372	401357	3,00
155	989	3916	0,936	0,000226	1,566	0,432	0,371	458335	3,57
160	986	3929	0,939	0,000218	1,527	0,431	0,371	521621	4,20
165	984	3943	0,942	0,000210	1,492	0,430	0,370	591705	4,90
170	981	3956	0,945	0,000204	1,460	0,430	0,370	669097	5,68
175	979	3969	0,948	0,000197	1,431	0,429	0,369	754328	6,53



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7.2. SAFETY DATA SHEET OF GLYCOL



SAFETY DATA SHEET

Creation Date 02-Feb-2010

Revision Date 17-Jan-2018

Revision Number 4

1. Identification

Product Name Ethylene glycol
Cat No. : E177-4; E177-20
CAS-No 107-21-1
Synonyms Monoethylene glycol; 1,2-Ethanediol
Recommended Use Laboratory chemicals.
Uses advised against Not for food, drug, pesticide or biocidal product use

Details of the supplier of the safety data sheet

Company
 Fisher Scientific
 One Reagent Lane
 Fair Lawn, NJ 07410
 Tel: (201) 796-7100

Emergency Telephone Number
 CHEMTREC®, Inside the USA: 800-424-9300
 CHEMTREC®, Outside the USA: 001-703-527-3887

2. Hazard(s) identification

Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Acute oral toxicity	Category 4
Specific target organ toxicity (single exposure)	Category 3
Target Organs - Central nervous system (CNS).	
Specific target organ toxicity - (repeated exposure)	Category 2
Target Organs - Kidney, Liver.	

Label Elements

Signal Word
 Warning

Hazard Statements
 Harmful if swallowed
 May cause drowsiness or dizziness
 May cause damage to organs through prolonged or repeated exposure



Precautionary Statements

Prevention

Wash face, hands and any exposed skin thoroughly after handling
Do not eat, drink or smoke when using this product
Do not breathe dust/fume/gas/mist/vapors/spray
Use only outdoors or in a well-ventilated area

Response

Get medical attention/advice if you feel unwell

Inhalation

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing
Call a POISON CENTER or doctor/physician if you feel unwell

Ingestion

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell
Rinse mouth

Storage

Store in a well-ventilated place. Keep container tightly closed
Store locked up

Disposal

Dispose of contents/container to an approved waste disposal plant
Hazards not otherwise classified (HNOC)

WARNING. Reproductive Harm - <https://www.p85warnings.ca.gov/>.

3. Composition/Information on Ingredients

Component	CAS-No	Weight %
Ethylene glycol	107-21-1	>95

4. First-aid measures

Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.
Inhalation	Move to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Get medical attention immediately if symptoms occur. If not breathing, give artificial respiration.
Ingestion	Do not induce vomiting. Call a physician or Poison Control Center immediately.
Most important symptoms and effects	Breathing difficulties.
Notes to Physician	Treat symptomatically

5. Fire-fighting measures

Suitable Extinguishing Media	Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.
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Ethylene glycol

Revision Date 17-Jan-2018

Unsuitable Extinguishing Media	No information available
Flash Point	111 °C / 231.8 °F
Method -	DIN 51758
Autoignition Temperature	413 °C / 775.4 °F
Explosion Limits	
Upper	15.30 vol %
Lower	3.20 vol %
Sensitivity to Mechanical Impact	No information available
Sensitivity to Static Discharge	No information available

Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

Hazardous Combustion Products

Carbon monoxide (CO) Carbon dioxide (CO₂)

Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

NFPA

Health
2

Flammability
1

Instability
1

Physical hazards
N/A

6. Accidental release measures

Personal Precautions	Ensure adequate ventilation. Use personal protective equipment.
Environmental Precautions	Should not be released into the environment. See Section 12 for additional ecological information.

Methods for Containment and Clean Up Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

7. Handling and storage

Handling	Wear personal protective equipment. Ensure adequate ventilation. Do not breathe vapors or spray mist. Avoid contact with skin, eyes and clothing.
Storage	Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat and sources of ignition.

8. Exposure controls / personal protection

Exposure Guidelines

Component	ACGIH TLV	OSHA PEL	NIOSH IDLH	Mexico OEL (TWA)
Ethylene glycol	TWA: 25 ppm STEL: 50 ppm STEL: 10 mg/m ³	(Vacated) Ceiling: 50 ppm (Vacated) Ceiling: 125 mg/m ³		Ceiling: 100 mg/m ³

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

Engineering Measures	Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.
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Personal Protective Equipment

Eye/face Protection	Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.
Skin and body protection	Wear appropriate protective gloves and clothing to prevent skin exposure.
Respiratory Protection	Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.
Hygiene Measures	Handle in accordance with good industrial hygiene and safety practice.

9. Physical and chemical properties

Physical State	Viscous liquid Liquid
Appearance	Colorless
Odor	Odorless
Odor Threshold	No information available
pH	5.5-7.5 50% aq. sol
Melting Point/Range	-13 °C / 8.6 °F
Boiling Point/Range	198 - 198 °C / 384.8 - 384.8 °F @ 760 mmHg
Flash Point	111 °C / 231.8 °F
Method -	DIN 51758
Evaporation Rate	No information available
Flammability (solid,gas)	Not applicable
Flammability or explosive limits	
Upper	15.30 vol %
Lower	3.20 vol %
Vapor Pressure	0.12 mmHg @ 20 °C
Vapor Density	2.14 (Air = 1.0)
Specific Gravity	1.113
Solubility	miscible
Partition coefficient; n-octanol/water	No data available
Autoignition Temperature	413 °C / 775.4 °F
Decomposition Temperature	> 500°C
Viscosity	21 cP (20°C)
Molecular Formula	C2 H8 O2
Molecular Weight	62.06

10. Stability and reactivity

Reactive Hazard	None known, based on information available
Stability	Hygroscopic.
Conditions to Avoid	Incompatible products. Excess heat. Exposure to moist air or water.
Incompatible Materials	Strong oxidizing agents, Strong acids, Strong bases, Aldehydes
Hazardous Decomposition Products	Carbon monoxide (CO), Carbon dioxide (CO ₂)
Hazardous Polymerization	Hazardous polymerization does not occur.
Hazardous Reactions	None under normal processing.

11. Toxicological information

Acute Toxicity

Ethylene glycol

Revision Date 17-Jan-2018

Product Information

Component Information

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Ethylene glycol	7712 mg/kg (Rat)	9530 µL/kg (Rabbit) 10600 mg/kg (Rat)	Not listed

Toxicologically Synergistic Products No information available

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Irritation May cause eye, skin, and respiratory tract irritation

Sensitization No information available

Carcinogenicity The table below indicates whether each agency has listed any ingredient as a carcinogen.

Component	CAS-No	IARC	NTP	ACGIH	OSHA	Mexico
Ethylene glycol	107-21-1	Not listed	Not listed	Not listed	Not listed	Not listed

Mutagenic Effects No information available

Reproductive Effects No information available.

Developmental Effects No information available.

Teratogenicity No information available.

STOT - single exposure Central nervous system (CNS)

STOT - repeated exposure Kidney Liver

Aspiration hazard No information available

Symptoms / effects, both acute and delayed No information available

Endocrine Disruptor Information No information available

Other Adverse Effects The toxicological properties have not been fully investigated.

12. Ecological information

Ecotoxicity

Do not empty into drains. .

Component	Freshwater Algae	Freshwater Fish	Microtox	Water Flea
Ethylene glycol	EC50: 8500 - 13000 mg/L, 96h (Pseudokirchneriella subcapitata)	LC50: = 16000 mg/L, 96h static (Poecilia reticulata) LC50: 40000 - 60000 mg/L, 96h static (Pimephales promelas) LC50: = 40761 mg/L, 96h static (Oncorhynchus mykiss) LC50: = 41000 mg/L, 96h (Oncorhynchus mykiss) LC50: 14 - 18 mL/L, 96h static (Oncorhynchus mykiss) LC50: = 27540 mg/L, 96h static (Lepomis macrochirus)	Not listed	EC50: = 46300 mg/L, 48h (Daphnia magna)

Persistence and Degradability Persistence is unlikely

Bioaccumulation/ Accumulation No information available.

Ethylene glycol

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Mobility

Will likely be mobile in the environment due to its water solubility.

Component	log Pow
Ethylene glycol	-1.93

13. Disposal considerations

Waste Disposal Methods Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

14. Transport information

DOT	Not regulated
TDG	Not regulated
IATA	Not regulated
IMDG/IMO	Not regulated

15. Regulatory information

All of the components in the product are on the following Inventory lists: X = listed

International Inventories

Component	TSCA	DSL	NDSL	EINECS	ELINCS	NLP	PICCS	ENCS	AICS	IECSC	KECL
Ethylene glycol	X	X	-	203-473-3	-		X	X	X	X	X

Legend:

X - Listed

E - Indicates a substance that is the subject of a Section 5(e) Consent order under TSCA.

F - Indicates a substance that is the subject of a Section 5(f) Rule under TSCA.

N - Indicates a polymeric substance containing no free-radical initiator in its inventory name but is considered to cover the designated polymer made with any free-radical initiator regardless of the amount used.

P - Indicates a commenced PMN substance

R - Indicates a substance that is the subject of a Section 6 risk management rule under TSCA.

S - Indicates a substance that is identified in a proposed or final Significant New Use Rule

T - Indicates a substance that is the subject of a Section 4 test rule under TSCA.

XU - Indicates a substance exempt from reporting under the Inventory Update Rule, i.e. Partial Updating of the TSCA Inventory Data Base Production and Site Reports (40 CFR 710(B)).

Y1 - Indicates an exempt polymer that has a number-average molecular weight of 1,000 or greater.

Y2 - Indicates an exempt polymer that is a polyester and is made only from reactants included in a specified list of low concern reactants that comprises one of the eligibility criteria for the exemption rule.

U.S. Federal Regulations

TSCA 12(b) Not applicable

SARA 313

Component	CAS-No	Weight %	SARA 313 - Threshold Values %
Ethylene glycol	107-21-1	>95	1.0

SARA 311/312 Hazard Categories See section 2 for more information

CWA (Clean Water Act) Not applicable

Clean Air Act

Component	HAPS Data	Class 1 Ozone Depletors	Class 2 Ozone Depletors
Ethylene glycol	X		-

OSHA Occupational Safety and Health Administration
Not applicable

Ethylene glycol

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CERCLA This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

Component	Hazardous Substances RQs	CERCLA EHS RQs
Ethylene glycol	5000 lb	-

California Proposition 65 This product does not contain any Proposition 65 chemicals

Component	CAS-No	California Prop. 65	Prop 65 NSRL	Category
Ethylene glycol	107-21-1	Developmental	-	Developmental

U. S. State Right-to-Know Regulations

Component	Massachusetts	New Jersey	Pennsylvania	Illinois	Rhode Island
Ethylene glycol	X	X	X	X	-

U. S. Department of Transportation

Reportable Quantity (RQ): Y
DOT Marine Pollutant N
DOT Severe Marine Pollutant N

U. S. Department of Homeland Security
This product does not contain any DHS chemicals.

Other International Regulations

Mexico - Grade Slight risk, Grade 1

16. Other information

Prepared By Regulatory Affairs
Thermo Fisher Scientific
Email: EMSDS.RA@thermofisher.com

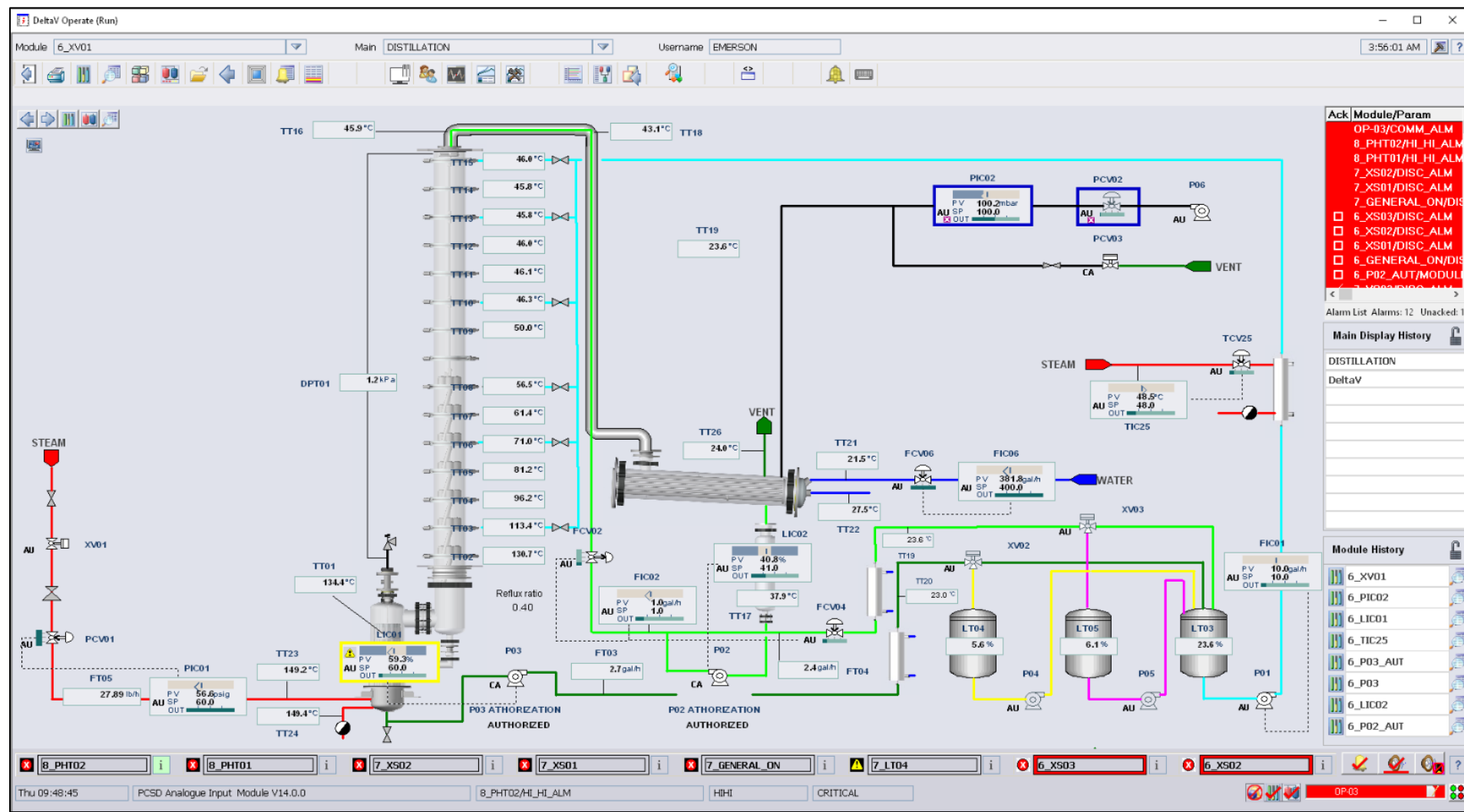
Creation Date 02-Feb-2010
Revision Date 17-Jan-2018
Print Date 17-Jan-2018
Revision Summary This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of SDS

7.3. CASE STUDY N°1: 100 MBAR (1.45 PSI) CONTINUOUS DISTILLATION WITH A 30% GLYCOL FEED MIXTURE (TEMPERTURE IN °C)



7.4 CASE STUDY N°2: 100 MBAR (1.45 PSI) CONTINUOUS DISTILLATION WITH A 30% GLYCOL FEED MISTURE (TEMPERATURE IN °F)

